

Untitled hardware project

Class vertical_farm	Feasibility feasible	BoM Readiness Ready	Score 7.7
------------------------	-------------------------	------------------------	--------------

Executive Summary

Metric	Value	Unit	Confidence	Notes
Annual leafy-green yield	1800	kg/year	low	Class-pack estimate for compact indoor leafy-green unit.
Estimated CAPEX	0	GBP	low	No CAPEX claim admitted yet; supplier-backed BoM evidence is required before costs appear.
Estimated annual OPEX	0	GBP/year	low	No OPEX claim admitted yet because CAPEX has no sourced BoM basis.

Brief Intake Gate

brief_ready_for_architecture

6/6 intake areas pass; 0 review, 0 blocked. Class vertical_farm (high); requirements 3; numeric facts 3; requirement trace 3/3 covered; scratch grammar yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_text	pass	1	141 non-whitespace character(s) in the source brief.	none	Brief text is present.
requirement_quantification	pass	1	3 quantified requirement(s), 3 numeric fact(s).	none	Quantified requirements are sufficient for a first architecture pass.
product_class_selection	pass	1	Selected vertical_farm with high confidence.	none	Product class selection is strong enough for scratch architecture.
scratch_design_support	pass	1	vertical_farm has a deep scratch architecture grammar.	none	Use scratch universal architecture flow for this brief.
requirement_trace_seed	pass	1	3 covered, 0 partial, 0 uncovered requirement trace row(s).	none	Extracted requirements have trace seeds into the architecture.
assumption_boundary	pass	1	1 brief assumption note(s); no-quantified-default path: no.	none	Assumption boundary is explicit enough for first-pass architecture.

Brief Clarification Plan

no_clarification_needed

No clarification questions are needed before scratch architecture generation.

Architecture Admission Gate

architecture_generation_admitted

6/6 admission areas pass; 0 review, 0 blocked. Can use for review: yes; can proceed to BoM: yes. Source scratch_universal_architecture; next action: none.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_intake_clearance	pass	1	6/6 intake areas pass; class vertical_farm (high); 3 quantified requirement(s).	none	Brief intake is strong enough to admit scratch architecture generation.
clarification_clearance	pass	1	0 clarification question(s): 0 required, 0 recommended, 0 architecture-blocking.	none	No brief clarification is needed before architecture generation.
class_grammar_clearance	pass	1	Product class vertical_farm; scratch universal grammar supported: yes.	none	Route this class through the scratch universal module grammar.
scratch_stage_clearance	pass	1	Architecture stage passed; source scratch_universal_architecture; modules 9; submodules 28; components 122.	none	Architecture was produced by the scratch universal architecture stage.
architecture_readiness_boundary	pass	1	Architecture ready for BoM: yes; blocking issues 0; required links 3/3.	none	Deterministic architecture readiness checks passed; sourcing work may begin only with explicit evidence.
bom_provenance_hold	pass	1	0 admitted sourcing evidence row(s), 0 source-backed BoM claim(s), 0 provenance violation(s).	none	BoM fields remain behind the explicit sourcing-evidence boundary.

Stage Integrity Gate

stage_trace_accepted

5/5 stage-integrity areas pass; 0 review, 0 blocked. Stages 8/8; ordered: yes; architecture source scratch_universal_architecture; admitted priced lines 0; provenance violations 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
stage_sequence	pass	1	8/8 expected compiler stages are present in order.	none	Stage sequence matches the scratch compiler contract.
stage_payloads	pass	1	8/8 stages have metrics; 8/8 have evidence or explicit limitation; 8/8 have limitations.	none	Every stage carries metrics, evidence or explicit limitation, and limitations.
scratch_architecture_source	pass	1	Product class vertical_farm; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Architecture came from the scratch universal grammar.
architecture_before_sourcing	pass	1	Architecture ready: yes; readiness stage passed; sourcing stage warning.	none	Sourcing stage is downstream of architecture readiness and has a consistent status.
sourcing_provenance_boundary	pass	1	0 admitted source evidence row(s), 0 admitted priced line(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN and cost fields stay behind admitted sourcing evidence.

Scratch Lineage Gate

scratch_lineage_clean

6/6 lineage areas pass; 0 review, 0 blocked. Architecture source scratch_universal_architecture; design refs 268; source refs 0; forbidden refs 0; forbidden stage mentions 0; chain-v2 benchmark no (prototype internal depth target).

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_source	pass	1	Product class vertical_farm; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Design content came from the scratch universal architecture stage.
design_provenance_refs	pass	1	518 provenance ref(s); 268 design/architecture ref(s); 0 invalid kind(s); 0 forbidden ref(s).	none	All design provenance references are in the allowed scratch provenance vocabulary.
bom_candidate_provenance	pass	1	122 candidate BoM line(s); 0 admitted source evidence row(s); 0 unprovenanced supplier/manufacturer/MPN/cost claim(s).	none	Candidate BoM fields remain unpriced or explicitly source-backed.
source_evidence_boundary	pass	1	0 source provenance ref(s); 0 outside admitted evidence boundary; 0 inside architecture content.	none	Source refs appear only in admitted source evidence, source ledger entries or matching source-backed BoM lines.
chain_v2_quarantine	pass	1	0 forbidden provenance ref(s); 0 forbidden stage-trace mention(s).	none	No chain-v2/state.json/modifier lineage appears inside generated design content or stage traces.
benchmark_isolation	pass	1	Benchmark source: prototype internal depth target; chain benchmark used: no; policy: Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.	none	Benchmark does not use chain-v2 input.

Architecture Freeze Gate

architecture_freeze_review_required

2/6 freeze areas pass; 4 review, 0 blocked. Modules/submodules/components 9/28/122. Structurally ready for sourcing yes; independent review accepted no. Module engineering module_engineering_review_required; interfaces interface_review_ready; requirements coverage_review_ready; next action: Treat the architecture as review-ready until module review questions and critical sourcing blocks close..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
admission_and_lineage	pass	1	Architecture admission architecture_generation_admitted; stage integrity stage_trace_accepted; scratch lineage scratch_lineage_clean.	none	Architecture was admitted from the scratch flow with clean stage and lineage evidence.
module_structure	review	0.76	Component allocation allocation_ready; module engineering module_engineering_review_required; 0/9 modules pass; 5 modules still have critical sourcing blocks.	none	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.
interface_contracts	review	1	Interface verification interface_review_ready; 3/3 required interfaces have carriers; 0 accepted.	none	Collect or accept interface-review evidence before calling the architecture fully frozen.
requirement_coverage	review	1	Requirement coverage coverage_review_ready; 3/3 requirements structurally ready or accepted; accepted ratio 0.	none	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.
engineering_review_state	review	0.67	Pre-BoM engineering engineering_review_ready; 4/6 areas pass; calculations needing review 1; assumptions needing review 16.	none	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.
sourcing_boundary	pass	1	122 candidate BoM lines; 0 admitted sourcing evidence rows; 0 provenance violation(s); 32 critical missing source claim(s).	none	Sourcing and BoM claims remain behind explicit admitted source evidence.

Architecture Freeze Closure Plan

architecture_freeze_review_required

4 closure row(s): 4 ready, 0 blocked. Queues S/E/V/R 1/1/2/0; next row freeze-close:1:module_structure. Structurally ready for sourcing yes; independent review accepted no.

#	Area	Queue	Priority	Status	Required Evidence	Acceptance Criteria	Input Artifacts
1	module_structure	sourcing_intake	major	ready	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.	Module engineering verdict is module_engineering_ready. No module has blocked allocation or unresolved critical sourcing blockers. Every module review question is accepted, revised or explicitly deferred.	*.module-engineering-gate.csv; *.component-allocation-gate.csv; *.engineering-review-pack.csv; *.sourcing-intake-template.csv
2	interface_contracts	verification_intake	minor	ready	Collect or accept interface-review evidence before calling the architecture fully frozen.	Interface verification verdict is accepted_interfaces. Every required interface has endpoint carriers and accepted reviewer evidence.	*.interface-contracts.json; *.interface-verification-gate.csv; *.verification-intake-template.csv
3	requirement_coverage	verification_intake	minor	ready	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.	Requirement coverage verdict is accepted_evidence. Every parsed requirement has architecture links and accepted verification evidence.	*.requirement-coverage-gate.csv; *.engineering-assurance-matrix.csv; *.verification-intake-template.csv
4	engineering_review_state	engineering_review	major	ready	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.	Pre-BoM engineering verdict is engineering_accepted. Calculations, assumptions and verification activities are accepted or explicitly deferred.	*.pre-bom-engineering-gate.csv; *.engineering-calculations.csv; *.engineering-assumptions.csv; *.verification-plan.csv

component_candidates_ready_for_sourcing

7/7 candidate areas pass; 0 review, 0 blocked. Candidate lines 122; critical 8/8; worklist rows 122; duplicate groups 0; provenance violations 0; ready for sourcing yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
candidate_presence	pass	1	122 candidate BoM line(s), 122 distinct component identity key(s).	none	Component candidates exist for downstream sourcing review.
candidate_identity	pass	1	122/122 candidate line(s) have non-generic identity, description and role context.	none	Candidate identities are specific enough for first-pass sourcing worklists.
quantity_basis	pass	1	122/122 candidate line(s) have positive quantity and unit.	none	Candidate quantities have positive values and explicit units.
critical_candidate_coverage	pass	1	8/8 class-critical part(s) are represented by candidate lines; 8 critical BoM line(s).	none	Class-critical parts have candidate lines before sourcing begins.
duplicate_identity_review	pass	1	0 duplicate component identity group(s), 0 involving critical lines.	none	No duplicate component identities need sourcing review.
sourcing_worklist_alignment	pass	1	122/122 unpriced candidate line(s) appear in the sourcing worklist.	none	Every unpriced candidate line is visible in the sourcing worklist.
provenance_boundary	pass	1	0 source evidence row(s), 0 source-backed claim(s), 0 provenance violation(s).	none	Candidate sourcing fields remain clean unless admitted evidence supplies them.

Sourcing Authorization Gate

sourcing_authorized

6/6 sourcing-authorization areas pass; 0 review, 0 blocked. Critical intake 8/8; full intake 122/122; admitted evidence 0; rejected evidence 0; provenance violations 0; authorized yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_authorization	pass	1	architecture_generation_admitted; review yes, BoM yes.	none	Architecture admission permits evidence-gated sourcing work.
component_candidate_authorization	pass	1	component_candidates_ready_for_sourcing; 122 candidate line(s), 122 sourcing-worklist row(s), 0 provenance violation(s).	none	Component candidates are concrete enough for sourcing intake.
critical_intake_scope	pass	1	8/8 critical unpriced line(s) appear in the critical sourcing intake template.	none	Critical sourcing intake scope is complete.
full_worklist_scope	pass	1	122/122 total unpriced line(s) appear in the full sourcing intake template.	none	Full sourcing intake scope matches the unpriced worklist.
evidence_admission_boundary	pass	1	0 admitted source evidence row(s), 0 rejected source evidence row(s).	none	No rejected sourcing evidence is waiting for repair.
provenance_boundary	pass	1	0 source-backed BoM claim(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN, lead-time and cost fields remain behind admitted evidence.

BoM Admission Gate

candidate_bom_authorized

Display mode candidate_only. 2/7 BoM-admission areas pass; 5 review, 0 blocked. Critical priced 0/8; source-backed claims 0; provenance violations 0. Candidate BoM yes; priced BoM no; procurement use no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_authorization	pass	1	sourcing_authorized; authorized yes; 6/6 areas pass.	none	Sourcing is authorized, so the BoM may be shown as a candidate sourcing queue.
candidate_bom_scope	pass	1	122 candidate BoM line(s), 8 critical line(s).	none	Candidate BoM scope exists for sourcing and evidence collection.
source_admission_state	review	0	0 admitted source row(s), 0 rejected source row(s), 0/122 priced line(s).	none	No source rows are admitted; keep this as a candidate-only BoM.
critical_bom_admission	review	0	0/8 critical line(s) have admitted unit costs; 8 critical line(s) remain unpriced.	none	Start with critical sourcing intake before treating any BoM cost as admitted.
source_reference_quality	review	0	no_sourcing_evidence; 0/0 source-quality rows pass; 0 protocol fixture row(s).	none	No source evidence is present; keep the BoM candidate-only.
provenance_boundary	review	0.95	0 source-backed claim(s), 32 critical missing-source claim(s), 0 provenance violation(s).	none	Missing critical source claims are acceptable only in candidate-only mode; collect evidence before priced BoM use.
display_mode	review	0.25	Display mode will be candidate_only.	none	Show candidate BoM as a sourcing queue with supplier, MPN and cost fields empty.

Report Readiness Gate

architecture_review_ready

5/6 sections at or above 8. Mean score 7.67. Architecture ready: yes. Critical unpriced lines: 8. Verification accepted: 0/19; unaccepted: 19. Verification acceptance ratio: 0.

- bom score 1 is below 8.
- 8 critical BoM line(s) lack source-backed cost evidence.
- 5 verification activity/activities remain blocked.
- 0/19 non-sourcing verification activity/activities have accepted reviewer evidence.

Section	Score	Target	B/M/m Issues	Action
executive_summary	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
brief_requirements	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
design_modules	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
bom	1	below target	8/1/0	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates.
feasibility_notes	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
sources_references	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.

Pre-BoM Engineering Gate

engineering_review_ready

4/6 areas pass; 2 review, 0 blocked. Modules/submodules/components: 9/28/122. Calculations within envelope 1/2; calculation blockers 0. Assumption review rows 16; architecture blockers 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	9 modules, 28 submodules and 122 component words pass architecture readiness.	none	Architecture is structurally ready for engineering review.
component_allocation	pass	1	28/28 submodules ready; 8/8 critical parts allocated; 0 duplicate component groups.	none	Component candidates are allocated well enough for engineering review.
interface_verification	pass	1	3/3 required interfaces have endpoint carriers; 0 accepted with evidence.	none	Required interfaces are structurally ready for reviewer evidence.
requirement_coverage	pass	1	3/3 requirements are structurally ready or accepted; accepted evidence ratio 0.	none	Requirements are structurally ready for reviewer evidence.
calculation_envelope	review	0.5	1/2 calculations are within envelope; 1 need review; 0 block.	none	Attach reviewer evidence to calculations marked needs_review.
engineering_assumptions	review	0.27	6/22 assumptions are brief-supported or model-present; 16 need review/source evidence; 0 block architecture.	none	Review assumptions and keep BoM/source-required assumptions out of trusted claims until evidence is admitted.

BoM Costing Gate

costing_not_started

1/7 costing areas pass; 0 review, 6 blocked. Critical priced lines 0/8; unpriced critical lines 8. Source-backed claims 0; provenance violations 0; sourcing evidence 0/0 production-ready, 0 protocol. Source quality: no_sourcing_evidence (0/0 pass, 0 placeholder URL, 0 quote-anchored). BoM total £0; CAPEX £0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_admission	blocked	0	0/122 BoM lines have admitted pricing; rejected source records 0.	structure_containment_growing_rack_stack_growing_rack_structure: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. environmental_interface_horticultural_lighting_led_grow_lights: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. environmental_interface_canopy_air_mixing_air_circulation_fans: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. environmental_interface_co2_enrichment_co2_sensor_or_dosing: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. mass_fluid_transport_process_reservoir_plumbing_reservoir_and_plumbing_set: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 3 more	Admit source-backed supplier, manufacturer, MPN and unit-cost evidence before costing review.
critical_line_pricing	blocked	0	0/8 critical lines have admitted unit costs.	structure_containment_growing_rack_stack_growing_rack_structure: Growing rack structure environmental_interface_horticultural_lighting_led_grow_lights: LED grow lights environmental_interface_canopy_air_mixing_air_circulation_fans: Air circulation fans environmental_interface_co2_enrichment_co2_sensor_or_dosing: CO2 sensor	Price every critical line from admitted source evidence.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				or dosing mass_fluid_transpo rt_process_reservoi r_plumbing_reservo ir_and_plumbing_se t: Reservoir and plumbing set plus 3 more	
source_backed_pro venance	blocked	0.95	0 source-backed claims; 32 critical missing-source claims; 0 provenance violations.	structure_containm ent_growing_rack_s tack_growing_rack_ structure/supplier: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. structure_containm ent_growing_rack_s tack_growing_rack_ structure/manufactu rer: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. structure_containm ent_growing_rack_s tack_growing_rack_ structure/mpn: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. structure_containm ent_growing_rack_s tack_growing_rack_ structure/unit_cost_ gbp: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. environmental_inter face_horticultural_li ghting_led_grow_lig hts/supplier: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. plus 27 more	Keep supplier, manufacturer, MPN, unit cost and lead- time fields empty until source-backed evidence is admitted.
component_identity	pass	1	0 duplicate component identity group(s); 0 include critical lines.	none	Component identities are unique enough for line- level costing.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_evidence_authenticity	blocked	0	0/0 sourcing evidence rows are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Add sourcing evidence rows before BoM costing can start.
source_reference_quality	blocked	0	0/0 sourcing refs pass quality checks; 0 placeholder URL, 0 quote-anchored, 0 fresh timestamp.	none	Add source references before costing review.
cost_totals	blocked	0	BoM total 0 GBP; CAPEX 0 GBP; annual OPEX 0 GBP.	8 critical line(s) are unpriced, so totals are incomplete.	Do not use cost totals until every critical line has an admitted unit cost.

Evidence Acquisition Plan

32 missing-evidence row(s): 11 sourcing, 21 verification. Ready 29; blocked 0. Blocks BoM 11; blocks publish 32. Next row: acquire:4:assumption:metric:capex_gbp.

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
1	sourcing	blocker	open	readiness:2	supplier_catalogue_or_quote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
2	verification	blocker	open	readiness:3	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
3	verification	blocker	open	readiness:4	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
4	sourcing	blocker	ready_for_intake	assumption:metric:capex_gbp	supplier_catalogue_or_quote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						sourceGrade; evidence.ref; evidence.quote; retrievedAt		
5	sourcing	blocker	ready_for_intake	assumption:metric:opex_annual_gbp	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
6	sourcing	blocker	ready_for_intake	growing_rack_structure	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
7	sourcing	blocker	ready_for_intake	led_grow_lights	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.ref ; evidence.quote; retrievedAt		
8	sourcing	blocker	ready_for_in take	air_circulation_fans	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
9	sourcing	blocker	ready_for_in take	co2_sensor_or_dosing	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
10	sourcing	blocker	ready_for_in take	reservoir_and_plumbing_set	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.quote; retrievedAt		
11	sourcing	blocker	ready_for_in take	nutrient_pu mp	supplier_cat alogue_or_q uote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
12	sourcing	blocker	ready_for_in take	environment al_controller	supplier_cat alogue_or_q uote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
13	sourcing	blocker	ready_for_in take	rcd_isolator _panel	supplier_cat alogue_or_q uote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						ote; retrievedAt		
14	verification	major	ready_for_in take	design_review:structure_containm ent	named_engineering_review	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
15	verification	major	ready_for_in take	calculation:structure_c ontainment	named_engineering_review	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
16	verification	major	ready_for_in take	design_review:environ mental_inte rface	named_engineering_review	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
17	verification	major	ready_for_in take	calculation:environmental_interface	named_engineering_review	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
18	verification	major	ready_for_intake	design_review:mass_fluid_transport_process	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
19	verification	major	ready_for_intake	design_review:sensing_instrumentation	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
20	verification	major	ready_for_intake	design_review:control_compute_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
21	verification	major	ready_for_intake	design_review:power_distribution	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
22	verification	major	ready_for_intake	design_review:safety_protection	named_engineering_review	activityId; evidenceKind; reviewerNa	anonymous reviewer; unchecked model	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						me; verdict; evidenceRef; evidenceNote; reviewedAt	assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	
23	verification	major	ready_for_intake	design_review:maintenance_serviceability	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
24	verification	major	ready_for_intake	calculation:maintenance_serviceability	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
25	verification	major	ready_for_intake	design_review:hmi_ergonomics	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
26	verification	major	ready_for_intake	interface_review:control_compute_communication:environmental_interface:actuator_bus	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNo	anonymous reviewer; unchecked model assertion; test- fixture:// review; source-	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						te; reviewedAt	evidence claims that belong in sourcing intake	
27	verification	major	ready_for_in take	interface_re view:control _compute_c ommunicati on:mass_flu id_transport _process:ac tuator_bus	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
28	verification	major	ready_for_in take	interface_re view:enviro nmental_int erface:mass _fluid_trans port_proces s:sensor_bu s	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
29	verification	major	ready_for_in take	compliance _review:CE- UKCA	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
30	verification	major	ready_for_intake	compliance_review:WRAS	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
31	verification	major	ready_for_intake	risk_review:1	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
32	verification	major	ready_for_intake	risk_review:2	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

Document Trust Gate

evidence_blocked

1/7 trust areas pass; 3 review, 3 blocked. Architecture ready: yes. Claim gate: evidence_blocked. Reviewer evidence 0/19. Source-backed BoM claims 0; critical missing BoM claims 32; provenance violations 0. Evidence authenticity: no_evidence (0/0 production-ready, 0 protocol).

- bom: score 1 below 8
- architecture design has 0 blocked/source-required claim(s) and 37 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s).
- bom sourcing has 40 blocked/source-required claim(s) and 114 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 4 review-required claim(s).
- structure_containment_growing_rack_stack_growing_rack_structure/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_growing_rack_stack_growing_rack_structure/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_growing_rack_stack_growing_rack_structure/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

- structure_containment_growing_rack_stack_growing_rack_structure/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- environmental_interface_horticultural_lighting_led_grow_lights/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- environmental_interface_horticultural_lighting_led_grow_lights/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- environmental_interface_horticultural_lighting_led_grow_lights/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	9 modules, 28 submodules and 3 required links pass deterministic architecture gates.	none	No deterministic architecture blocker remains; continue with reviewer evidence before treating it as accepted.
section_scores	blocked	0.83	5/6 scored sections are at or above 8; mean 7.67.	bom: score 1 below 8	Repair below-target sections through the specific score-ledger actions.
claim_evidence	blocked	0.01	3/203 claims pass; 42 blocked/source-required, 158 review-required.	architecture design has 0 blocked/source-required claim(s) and 37 review-required claim(s). engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s). bom sourcing has 40 blocked/source-required claim(s) and 114 review-required claim(s). compliance risk has 0 blocked/source-required claim(s) and 4 review-required claim(s).	Accept, correct or reject generated module and submodule allocation claims through engineering review. Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence. Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims. Accept, correct or reject class-pack regulatory and risk claims through compliance/domain review.
requirement_assurance	review	0	0/3 requirements have accepted assurance; 3 need review, 0 blocked, 0 unlinked.	none	Collect reviewer evidence for requirement assurance rows before publication.
reviewer_evidence	review	0	0/19 non-sourcing verification activities accepted; 0 rejected, 0 deferred, 19 pending/blocked.	none	Collect accepted reviewer evidence for every non-sourcing verification activity.
bom_provenance	blocked	0.95	0 source-backed BoM claims, 32 critical missing-source claims, 0 provenance violations.	structure_containment_growing_rack_stack_growing_rack_stack_supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_growing_rack_stack_growing_rack_	Admit source-backed evidence for critical BoM supplier, manufacturer, MPN and cost claims.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				structure/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_growing_rack_stack_growing_rack_structure/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_growing_rack_stack_growing_rack_structure/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. environmental_interface_horticultural_lighting_led_grow_lights/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 27 more	

evidence_authenticity	review	0	0/0 evidence row(s) are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Admit source and reviewer evidence before trusting the document.
-----------------------	--------	---	--	------	--

Evidence Authenticity Gate

no_evidence

0/0 evidence rows are production-ready. Protocol fixtures 0; review-required 0; missing metadata 0. External URLs 0; internal references 0; local files 0; unknown references 0.

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
------	---------	-----------------	--------	-----	--------	-----------------

Evidence Replacement Plan

No evidence replacement rows remain for the current authenticity verdict: no_evidence.

Trust Repair Plan

6 package(s): 6 ready, 0 waiting. Next package: sourcing_intake. Closure rows 40; trust blockers 45. Trust verdict: evidence_blocked.

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
2	Admit Source-Backed BoM Claims	ready	claim_evidence; bom_provenance	11	36	Supplier, manufacturer, MPN, unit cost, lead time, retrieved-at timestamp and source reference for critical BoM lines. Rejected or duplicate component identities resolved before pricing shared items.	*.sourcing-intake-template.csv; *.sourcing-pack.csv; *.bom-provenance-manifest.csv; *.component-identity.csv	BoM provenance has zero critical missing-source claims. Claim Evidence Gate BoM sourcing area is no longer evidence_blocked.
3	Resolve Engineering Review Questions	ready	claim_evidence; requirement_assurance	7	4	Accepted/corrected/rejected answers to review-pack questions. Calculation and assumption decisions tied to reviewer evidence references.	*.engineering-review-pack.csv; *.engineering-assurance-matrix.csv; *.engineering-assumptions.csv; *.engineering-calculations.csv	Requirement assurance rows have accepted or explicitly deferred engineering review evidence. Generated architecture and calculation claims are accepted, corrected or rejected.
4	Admit Reviewer Verification Evidence	ready	reviewer_evidence	21	0	Accepted, rejected or deferred verification evidence for every non-source verification activity.	*.verification-intake-template.csv; *.verification-plan.csv; *.verification-ledger.csv	Every non-source verification activity has accepted reviewer evidence or an explicit defer decision.
5	Validate Evidence Authenticity	ready	evidence_authenticity	0	0	Production supplier/catalogue URLs for BoM source records. Named reviewer evidence references that are not protocol fixtures.	*.evidence-authenticity-gate.csv; *.sourcing-ledger.csv; *.verification-ledger.csv; *.claim-ledger.csv	Evidence Authenticity Gate verdict is production_ready. No protocol_fixture, unknown_reference or missing_metadata evidence

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
								rows remain.
6	Review Compliance And Risk Claims	ready	claim_evidence	0	4	Compliance -domain review of class-pack standards, hazards and mitigation claims.	*.claim-ledger.csv; *.engineering-review-pack.csv	Compliance and risk claims are accepted, corrected or rejected by a domain reviewer.
7	Repair Residual Section Scores	ready	section_scores	1	1	A rerun after evidence intake, plus score-ledger fixes for any section still below target.	*.score-ledger.csv; *.readiness-gate.csv; *.document-trust-gate.csv	All scored sections are at or above target and Document Trust Gate is no longer blocked by scoring.

Claim Ledger

203 claims: 3 brief-supplied, 0 reviewer-accepted, 0 source-backed, 3 calculated needing review, 155 generated needing review, 42 source-required, 0 blocked. Blocking: 42 BoM, 199 publishable.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	Environmental controller / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Environmental controller / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Environmental controller / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Environmental controller / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Air circulation fans / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Air circulation fans / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	Air circulation fans / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Air circulation fans / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	CO2 sensor or dosing / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	CO2 sensor or dosing / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	CO2 sensor or dosing / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	CO2 sensor or dosing / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								cost evidence before BoM can pass.
bom_source_field	source_required	assumption	LED grow lights / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	LED grow lights / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	LED grow lights / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	LED grow lights / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Nutrient pump / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Nutrient pump / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/ma

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Nutrient pump / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Nutrient pump / unit_cost_g bp	unit_cost_g bp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Reservoir and plumbing set / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Reservoir and plumbing set / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Reservoir and plumbing set / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Reservoir and	unit_cost_g bp is	none	yes	yes	Admit source-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
			plumbing set / unit_cost_g bp	required but not source-backed.				backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	RCD isolator panel / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	RCD isolator panel / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	RCD isolator panel / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	RCD isolator panel / unit_cost_g bp	unit_cost_g bp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Growing rack structure / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	Growing rack structure / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Growing rack structure / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Growing rack structure / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
component_candidate	source_required	generated_model	Environmental controller	Environmental controller is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_arm	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Air circulation fans	Air circulation fans is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_arm	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	CO2 sensor or dosing	CO2 sensor or dosing is an	model:scratch_universal_architectu	yes	yes	Admit source-backed

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				architecture -derived component candidate with quantity 1 candidate.	re.vertical_f arm			supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	LED grow lights	LED grow lights is an architecture -derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_f arm	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Nutrient pump	Nutrient pump is an architecture -derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_f arm	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Reservoir and plumbing set	Reservoir and plumbing set is an architecture -derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_f arm	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	RCD isolator panel	RCD isolator panel is an architecture -derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_f arm	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								sourcing intake.
component_candidate	source_required	generated_model	Growing rack structure	Growing rack structure is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
headline_metric	source_required	assumption	Estimated CAPEX	Estimated CAPEX = 0 GBP.	formula:capex=sum(bom.totalCostGbp); class_pack:vertical_farm.class_pack	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
headline_metric	source_required	assumption	Estimated annual OPEX	Estimated annual OPEX = 0 GBP/year.	formula:opex=capex*0.04	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
component_candidate	generated_needs_review	generated_model	analog input module	analog input module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	DIN-rail PLC	DIN-rail PLC is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	real-time clock module	real-time clock module is an architecture-derived component candidate with	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				quantity 1 candidate.				
component_candidate	generated_needs_review	generated_model	relay output module	relay output module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	alarm historian	alarm historian is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	data export module	data export module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	recipe database	recipe database is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	set-point scheduler	set-point scheduler is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	industrial Ethernet switch	industrial Ethernet switch is an architecture-derived component	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				candidate with quantity 1 candidate.				or separate installation instance.
component_candidate	generated_needs_review	generated_model	MQTT telemetry bridge	MQTT telemetry bridge is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	SMS alarm modem	SMS alarm modem is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	Wi-Fi gateway	Wi-Fi gateway is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	airflow baffle plate	airflow baffle plate is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	fan guard grille	fan guard grille is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	replaceable intake filter	replaceable intake filter is an architecture-derived	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				component candidate with quantity 1 candidate.				component or separate installation instance.
component_candidate	generated_need_review	generated_model	CO2 distribution tube	CO2 distribution tube is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_arm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_need_review	generated_model	CO2 solenoid valve	CO2 solenoid valve is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_arm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_need_review	generated_model	gas regulator	gas regulator is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_arm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_need_review	generated_model	NDIR CO2 sensor head	NDIR CO2 sensor head is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_arm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_need_review	generated_model	dimnable LED driver	dimnable LED driver is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_arm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	glare shield	glare shield is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_arm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	lighting suspension bracket	lighting suspension bracket is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_arm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	PPFD calibration target	PPFD calibration target is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_arm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	condensate pump	condensate pump is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_arm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	dehumidifier module	dehumidifier module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_arm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	heat exchanger coil	heat exchanger coil is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	humidity control relay	humidity control relay is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	supply air temperature probe	supply air temperature probe is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.vertical_farm	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Showing first 70 of 203 claims; export CSV for the full claim ledger.

Claim Evidence Gate

evidence_blocked

3/203 claims pass by being brief-supplied, source-backed or reviewer-accepted. 42 blocked/source-required claims, 158 review-required claims. Claim pass ratio 0.01. Blocking: 42 BoM, 199 publishable.

- architecture design has 0 blocked/source-required claim(s) and 37 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s).
- bom sourcing has 40 blocked/source-required claim(s) and 114 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 4 review-required claim(s).

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
brief_requirements	complete	3	3	0	0	0	0	0	1	No claim-evidence action required for this area.
architecture_design	review	37	0	37	0	0	0	37	0	Accept, correct or reject generated module and submodule allocation claims through engineering review.
engineering_math	blocked	5	0	3	2	2	2	4	0	Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence.
bom_sourcing	blocked	154	0	114	40	40	40	154	0	Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims.
compliance_risk	review	4	0	4	0	0	0	4	0	Accept, correct or reject class-pack regulatory and risk claims through compliance

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
										ce/domain review.

Evidence Gap Register

40 open evidence gaps: 14 blockers, 26 major, 0 minor. Closure paths: 11 sourcing intake, 21 verification intake, 7 engineering review, 0 architecture revision. Blocking rows: architecture 0, BoM 11, publishable 40.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
readiness_blocker	blocker	open	Report readiness gate	bom score 1 is below 8.	score_repair	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	8 critical BoM line(s) lack source-backed cost evidence.	sourcing_intake	bom; publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	5 verification activity/activities remain blocked.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	0/19 non-sourcing verification activity/activities have accepted reviewer evidence.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier,

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							manufacturer, MPN and cost evidence for critical BoM lines.
calculation_review	major	needs review	Indicative service-clearance footprint	4.2 m2 allows a provisional 25% service envelope around the parsed footprint.	engineering_review	publishable	Review maintenance access, nutrient tank access, airflow paths and electrical service clearances.
assumption_review	major	needs review	Annual leafy-green yield	Annual leafy-green yield uses class_pack_default with low confidence.	engineering_review	publishable	Review the formula, input values and operating envelope before treating this metric as a performance claim.
assumption_review	blocker	ready for intake	Estimated CAPEX	Estimated CAPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	blocker	ready for intake	Estimated annual OPEX	Estimated annual OPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	major	needs review	Grow Unit Footprint	Grow Unit Footprint currently evaluates as pass against 1 to 20 m2 for compact prototype grow units.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Wet/Electrical Separation	Wet/Electrical Separation currently evaluates as pass against washdown boundary, mains supply and protective-earth interfaces present.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
assumption_review	major	needs review	Climate/Nutrient/Control Closure	Climate/Nutrient/Control Closure currently evaluates as pass against environmental , nutrient transport, sensing and control modules all present.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	CE-UKCA	Electrical safety and UKCA/CE file is relevant to this product class in UK/EU.	engineering_review	publishable	Technical file, electrical risk assessment and declarations.
assumption_review	major	needs review	WRAS	Water fittings compliance is relevant to this product class in UK.	engineering_review	publishable	Approved fittings or evidence for water-contact components.
sourcing_evidence	blocker	ready for intake	Growing rack structure	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	LED grow lights	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Air circulation fans	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							sourcing intake.
sourcing_evidence	blocker	ready for intake	CO2 sensor or dosing	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Reservoir and plumbing set	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Nutrient pump	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Environmental controller	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	RCD isolator panel	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							ce through sourcing intake.
verification_evidence	major	ready for intake	Growing Structure And Containment	Review Growing Structure And Containment architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Growing Structure And Containment	Verify Growing Structure And Containment requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Light Climate And Gas Environment	Review Light Climate And Gas Environment architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Light Climate And Gas Environment	Verify Light Climate And Gas Environment requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Nutrient And Water Transport Process	Review Nutrient And Water Transport Process architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Crop And Process Instrumentation	Review Crop And Process Instrumentation architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Recipe Control And Communications	Review Recipe Control And Communications architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Wet-Zone Power Distribution	Review Wet-Zone Power Distribution	verification_intake	publishable	Collect reviewer evidence through the

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				architecture allocation			verification intake template.
verification_evidence	major	ready for intake	Food-Safe Safety And Protection	Review Food-Safe Safety And Protection architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Harvest Cleaning And Serviceability	Review Harvest Cleaning And Serviceability architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Harvest Cleaning And Serviceability	Verify Harvest Cleaning And Serviceability requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Operator HMI And Labelling	Review Operator HMI And Labelling architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Recipe Control And Communications -> Light Climate And Gas Environment	Verify actuator_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Recipe Control And Communications -> Nutrient And Water Transport Process	Verify actuator_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Light Climate And Gas Environment -> Nutrient And Water Transport Process	Verify sensor_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for CE-UKCA	verification_intake	publishable	Collect reviewer evidence through the

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for WRAS	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Crop loss from pump failure	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Electrical fault in humid grow area	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Closure Plan

4 active phases. Next phase: sourcing_intake. Rows by phase: architecture 0, sourcing 11, engineering 7, verification 21, score repair 1.

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
2	Admit Source-Backed BoM Evidence	11	11/0/0	bom; publishable	Critical BoM costs, suppliers, manufacturers and MPNs must be source-backed before cost claims are credible.	Every critical BoM line has admitted supplier, manufacturer, MPN, unit cost and source reference. BoM provenance manifest has zero critical missing-source claims.	assumption:metric:capex_gbp; assumption:metric:opex_annual_gbp; readiness:2; sourcing:air_circulation_fans; sourcing:co2_sensor_or_dosing; sourcing:environmental_controller; sourcing:growing_rack_structure; sourcing:led_grow_lights; plus 3 more
3	Complete Engineering Review	7	0/7/0	publishable	Deterministic arithmetic can expose claims, but engineering judgment must review envelopes, assumptions and operating cases.	All calculation and assumption review rows are accepted, revised or explicitly deferred with reviewer rationale.	assumption:compliance:CE-UKCA; assumption:compliance:WRAS; assumption:metric:headline_output; assumption:sanity:farm_footprint; assumption:sanity:farm_process_closure; assumption:sanity:farm_wet_electrical_separation; calculation:farm_service_clearance_area_m2
4	Accept Reviewer Evidence	21	2/19/0	publishable	Reviewer acceptance is tracked separately from deterministic scoring and must close before publication.	Every non-source verification activity is accepted, or explicitly deferred and removed from publication criteria.	readiness:3; readiness:4; verification:calculation:environmental_interface; verification:calculation:maintenance_serviceability; verification:calculation:structure_containment; verification:compliance_review:CE-UKCA; verification:co

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
							mpliance_review:WRAS; verification:design_review:control_compute_communication; plus 12 more
5	Repair Remaining Section Scores	1	1/0/0	publishable	After evidence intake, rerun scoring to remove residual deterministic section blockers.	Every scored report section is at or above target and readiness verdict is publishable.	readiness:1

Depth Benchmark

prototype internal depth target. Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.

3 meeting benchmark, 4 below benchmark, average comparable ratio 0.85.

Dimension	Scratch	Benchmark	Ratio	Status	Notes	Action
Functional modules	9 modules	9 modules	1	meets	Core architecture breadth.	Add missing functional modules only when the product class genuinely requires them.
Submodules	28 submodules	30 submodules	0.93	below	Submodule count is a rough proxy for engineering decomposition depth.	Expand thin modules with concrete sub-functions and carrier interfaces.
Component candidates	122 components	140 components	0.87	below	Component-word count is a rough proxy for BoM review surface, not sourcing quality.	Add justified component candidates where submodules still hide assemblies.
Interface link volume	33 links	30 links	1.1	meets	Compares link volume only; chain-v2 grammar links and scratch interface edges are not semantically identical.	Add interface contracts only where they clarify a real energy, material, signal, load or service path.
Admitted priced-line ratio	0 ratio	0.75 ratio	0	below	Prototype target expects most critical lines to become source-backed before publication.	Use sourcing intake to admit source-backed price/manufacturer/MPN evidence; do not copy benchmark estimates.
Sections scoring >=8	5 sections	6 sections	0.83	below	Readiness gate requires every scored section to reach the target.	Clear section-specific readiness actions, especially BoM evidence.
Verification activities	24 activities	20 activities	1.2	meets	Scratch-only evidence workflow depth; not compared to chain-v2 content.	Keep adding verification activities when new requirements, interfaces or standards are introduced.

Brief Requirements

Design a compact vertical farm for leafy greens using LED grow lights, hydroponic fertigation, CO2 monitoring, and a 2.4 m by 1.4 m envelope.

Requirement	Value	Unit
Envelope length	2.4	m
Envelope width	1.4	m
Footprint	3.36	m2

Requirement Traceability

Requirement	Status	Value	Architecture Coverage	Evaluators	Notes
Envelope length	covered	2.4 m	Growing Structure And Containment / Growing rack stack; Harvest Cleaning And Serviceability / Harvest access; Light Climate And Gas Environment / Horticultural lighting	sanity:farm_footprint; sanity:farm_wet_electrical_separation; sanity:farm_process_closure	Envelope dimensions are traced through rack structure, service access and environmental closure. Requirement has architecture coverage plus metric or engineering sanity coverage.
Envelope width	covered	1.4 m	Growing Structure And Containment / Growing rack stack; Harvest Cleaning And Serviceability / Harvest access; Light Climate And Gas Environment / Horticultural lighting	sanity:farm_footprint; sanity:farm_wet_electrical_separation; sanity:farm_process_closure	Envelope dimensions are traced through rack structure, service access and environmental closure. Requirement has architecture coverage plus metric or engineering sanity coverage.
Footprint	covered	3.36 m2	Growing Structure And Containment / Growing rack stack; Harvest Cleaning And Serviceability / Harvest access; Light Climate And Gas Environment / Horticultural lighting	sanity:farm_footprint; sanity:farm_wet_electrical_separation; sanity:farm_process_closure	Envelope dimensions are traced through rack structure, service access and environmental closure. Requirement has architecture coverage plus metric or engineering sanity coverage.

Requirement Coverage Gate

coverage_review_ready

3 requirement row(s): 0 accepted with evidence, 3 coverage-ready, 0 needing review, 0 blocked, 0 unlinked. Structural coverage ratio 1; accepted evidence ratio 0.

Requirement	Status	Present Signals	Missing Signals	Modules	Verification Activities	Required Action
Envelope length	review ready	architecture_link ; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	structure_containment; maintenance_serviceability; environmental_interface	design_review:structure_containment; calculation:structure_containment; design_review:environmental_interface; calculation:environmental_interface; design_review:maintenance_serviceability; calculation:maintenance_serviceability	Collect accepted reviewer evidence; structural requirement coverage is ready for review.
Envelope width	review ready	architecture_link ; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	structure_containment; maintenance_serviceability; environmental_interface	design_review:structure_containment; calculation:structure_containment; design_review:environmental_interface; calculation:environmental_interface; design_review:maintenance_serviceability; calculation:maintenance_serviceability	Collect accepted reviewer evidence; structural requirement coverage is ready for review.
Footprint	review ready	architecture_link ; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	structure_containment; maintenance_serviceability; environmental_interface	design_review:structure_containment; calculation:structure_containment; design_review:environmental_interface; calculation:environmental_interface; design_review:maintenance_serviceability; calculation:maintenance_serviceability	Collect accepted reviewer evidence; structural requirement coverage is ready for review.

Engineering Assurance Matrix

3 requirements: 0 accepted, 0 ready, 3 needing review, 0 blocked, 0 unlinked. 3 have calculation links, 3 have review questions, 0 have accepted verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
Envelope length	needs review	covered	Growing Structure And Containment; Harvest Cleaning And Serviceability; Light Climate And Gas Environment	farm_footprint_check_m2	assumption_resolution: component: environmental_interface_canopy_air_mixing_air_circulation_fans; assumption_resolution: component: environmental_interface_co2_enrichment_co2_sensor_or_dosing; assumption_resolution: component: environmental_interface_horticultural_lighting_led_grow_lights; assumption_resolution: component: structure_containment_growing_rack_stack_growing_rack_structure; assumption_resolution: sanity:farm_footprint; plus 19 more	pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.

Envelope width	needs review	covered	Growing Structure And Containment; Harvest Cleaning And Serviceability; Light Climate And Gas Environment	farm_footprint_check_m2	assumption_resolution: component: environmental_interface_canopy_air_mixing_air_circulation_fans; assumption_resolution: component: environmental_interface_co2_enrichment_co2_sensor_or_dosing; assumption_resolution: component:	pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.
----------------	--------------	---------	---	-------------------------	--	---	------	--

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
					environmental_interface_horticultural_lighting_led_grow_lights; assumption_resolution: component: structure_containment_growing_rack_stack_growing_rack_structure; assumption_resolution: sanity:farm_footprint; plus 19 more			
Footprint	needs review	covered	Growing Structure And Containment; Harvest Cleaning And Serviceability; Light Climate And Gas Environment	farm_footprint_check_m2; farm_service_clearance_area_m2	assumption_resolution: component: environmental_interface_canopy_air_mixing_air_circulation_fans; assumption_resolution: component: environmental_interface_co2_enrichment_co2_sensor_or_dosing; assumption_resolution: component: environmental_interface_horticultural_lighting_led_grow_lights; assumption_resolution: component: structure_containment_growing_rack_stack_growing_rack_structure; assumption_resolution: sanity:farm_footprint; plus 19 more	pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.

Compiler Stage Trace

Stage	Status	Summary	Metrics	Evidence	Limitations
Brief Parsing	passed	Extracted quantified requirements from the user brief before design generation.	requirement_count: 3; numeric_fact_count : 3; requirement_trace_covered: 3; requirement_trace_partial: 0; requirement_trace_uncovered: 0	envelope_length_m: 2.4 m; envelope_width_m: 1.4 m; footprint_m2: 3.36 m2; envelope_length_m: covered; envelope_width_m: covered; footprint_m2: covered	Parsing is deterministic and currently recognises only a small set of hardware requirement patterns.
Product-Class Selection	passed	Selected vertical_farm with high confidence from brief keywords.	selected_class: vertical_farm; confidence: high; keyword_scores: vertical_farm:9	vertical_farm: 9	Classifier is keyword based; ambiguous multi-domain briefs need an LLM or richer taxonomy pass.
Universal Module Architecture	passed	Built a 9-module vertical_farm architecture from the scratch universal architecture grammar.	module_count: 9; architecture_source : scratch_universal_architecture	Growing Structure And Containment; Light Climate And Gas Environment; Nutrient And Water Transport Process; Crop And Process Instrumentation; Recipe Control And Communications; Wet-Zone Power Distribution; plus 3 more	The scratch grammar is hand-authored and deterministic; it still needs independent engineering review.
Submodule Expansion	passed	Expanded each module into engineering submodules with local purpose, interfaces and component candidates.	submodule_count: 28; average_submodules_per_module: 3.1	structure_containment.growing_rack_stack; structure_containment.crop_tray_carriers; structure_containment.enclosure_panels; environmental_interface.horticultural_lighting; environmental_interface.canopy_air_mixing; environmental_interface.humidity_temperature_control; plus 22 more	Submodule completeness is judged by deterministic coverage checks, not by a physics simulator.
Interface Graph	passed	Checked class-required module-to-module interfaces before allowing BoM review.	required_links: 3; present_links: 3; missing_links: 0; graph_nodes: 37; graph_edges: 61; shared_interface_edges: 30; missing_required_graph_edges: 0	control_compute_communication -> environmental_interface via actuator_bus: present; control_compute_communication -> mass_fluid_transpo	Interfaces prove allocation coverage only; they do not yet validate ratings, thermal margins or dynamic behaviour.

Stage	Status	Summary	Metrics	Evidence	Limitations
				rt_process via actuator_bus: present; environmental_interface -> mass_fluid_transport_process via sensor_bus: present	
Component Candidates	passed	Converted allocated component words into a candidate BoM without supplier prices.	candidate_lines: 122; critical_candidate_lines: 8; admitted_price_lines: 0	growing_rack_structure: Growing rack structure; aluminium_extrusion_upright: aluminium extrusion upright; cross_brace_set: cross brace set; adjustable_levelling_foot: adjustable levelling foot; tray_slide_rail: tray slide rail; food_safe_grow_tray: food-safe grow tray; plus 24 more	Showing first 30 of 122 component candidates in the stage trace.
Architecture Readiness Gate	passed	Architecture validators found no blocker or major issue, so candidate BoM review can start.	ready_for_bom: true; blocking_issue_count: 0; sanity_pass: 3; sanity_warn: 0; sanity_fail: 0	pass Grow Unit Footprint: 3.36 m2; pass Wet/Electrical Separation: present; pass Climate/Nutrient/Control Closure: present	Passing this gate means deterministic architecture checks passed; it is not a certified engineering sign-off.
Sourcing And BoM Admission	warning	No supplier, manufacturer, part number or cost claims have been admitted because no source-backed evidence was provided.	admission_status: not_started; candidate_lines: 122; admitted_lines: 0; unpriced_lines: 122; unpriced_critical_lines: 8; rejected_source_records: 0	none	BoM cost remains intentionally blocked until 8 critical candidate line(s) have explicit source evidence.

Engineering Architecture Review

Ready for BoM review

9 modules, 28 sub-modules, 122 allocated component words.

From	To	Interface	Status	Engineering Reason
control_compute_communication	environmental_interface	actuator_bus	OK	Farm controller must command lighting, CO2 dosing and airflow.
control_compute_communication	mass_fluid_transport_process	actuator_bus	OK	Farm controller must command fertigation pumps and nutrient circulation.
environmental_interface	mass_fluid_transport_process	sensor_bus	OK	Climate and fertigation state must be observable by the same control layer.

Interface Graph Summary

Module Nodes	Submodule Nodes	Shared Interface Edges	Required Edges	Missing Required Edges
9	28	30	3	0

Interface Contract Matrix

3/3 required contracts present. 12 shared interface families, 1 local-only declarations.

Interface	From Module	From Carrier	To Module	To Carrier	Status	Reason	Notes
actuator_bus	Recipe Control And Communications	Environmental controller	Light Climate And Gas Environment	Canopy air mixing; Humidity and temperature control; CO2 enrichment	present	Farm controller must command lighting, CO2 dosing and airflow.	Both endpoints expose the required interface through at least one submodule carrier.
actuator_bus	Recipe Control And Communications	Environmental controller	Nutrient And Water Transport Process	Fertigation pumping; Nutrient dosing	present	Farm controller must command fertigation pumps and nutrient circulation.	Both endpoints expose the required interface through at least one submodule carrier.
sensor_bus	Light Climate And Gas Environment	Humidity and temperature control; CO2 enrichment	Nutrient And Water Transport Process	Fertigation pumping; Nutrient dosing	present	Climate and fertigation state must be observable by the same control layer.	Both endpoints expose the required interface through at least one submodule carrier.

Local-only declarations to review later: protective_earth (Wet-Zone Power Distribution).

Interface Verification Gate

interface_review_ready

3 required interface row(s): 0 accepted, 3 review-ready, 0 pending evidence, 0 deferred, 0 blocked. Structural pass ratio 1; accepted evidence ratio 0.

Interface	From	To	Carrier Status	Verification Activity	Ledger	Status	Required Action
actuator_bus	Recipe Control And Communications	Light Climate And Gas Environment	both_endpoints_carried	interface_review:control_compute_communication:environmental_interface:actuator_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
actuator_bus	Recipe Control And Communications	Nutrient And Water Transport Process	both_endpoints_carried	interface_review:control_compute_communication:mass_fluid_transport_process:actuator_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
sensor_bus	Light Climate And Gas Environment	Nutrient And Water Transport Process	both_endpoints_carried	interface_review:environmental_interface:mass_fluid_transport_process:sensor_bus	pending	review_ready	Collect named interface-review evidence through verification intake.

Component Allocation Gate

allocation_ready

28/28 submodules ready; 0 review, 0 blocked. Component words 122; critical parts 8/8; duplicate groups 0. Allocation ratio 1; critical allocation ratio 1.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Growing Structure And Containment	Growing rack stack	ready	5	tray_support; service_accesses	growing_rack_structure	none	Submodule has component candidates and no allocation blocker.
Growing Structure And Containment	Crop tray carriers	ready	5	tray_support; nutrient_loop	none	none	Submodule has component candidates and no allocation blocker.
Growing Structure And Containment	Enclosure panels	ready	4	washdown_boundary; airflow	none	none	Submodule has component candidates and no allocation blocker.
Light Climate And Gas Environment	Horticultural lighting	ready	5	lighting_control; mains_supply; service_accesses	led_grow_lights	none	Submodule has component candidates and no allocation blocker.
Light Climate And Gas Environment	Canopy air mixing	ready	4	airflow; actuator_bus	air_circulation_fans	none	Submodule has component candidates and no allocation blocker.
Light Climate And Gas Environment	Humidity and temperature control	ready	5	airflow; sensor_bus; actuator_bus	none	none	Submodule has component candidates and no allocation blocker.
Light Climate And Gas Environment	CO2 enrichment	ready	5	airflow; sensor_bus; actuator_bus	co2_sensor_or_dosing	none	Submodule has component candidates and no allocation blocker.
Nutrient And Water Transport Process	Reservoir and plumbing	ready	5	nutrient_loop; service_accesses	reservoir_and_plumbing_set	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Nutrient And Water Transport Process	Fertigation pumping	ready	5	nutrient_loop; actuator_bus; sensor_bus	nutrient_pump	none	Submodule has component candidates and no allocation blocker.
Nutrient And Water Transport Process	Nutrient dosing	ready	5	nutrient_loop; actuator_bus; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Nutrient And Water Transport Process	Filtration and sanitation	ready	4	nutrient_loop; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Crop And Process Instrumentation	Climate sensors	ready	4	sensor_bus; airflow	none	none	Submodule has component candidates and no allocation blocker.
Crop And Process Instrumentation	Nutrient sensors	ready	4	sensor_bus; nutrient_loop	none	none	Submodule has component candidates and no allocation blocker.
Crop And Process Instrumentation	Crop observation	ready	4	video_link; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Recipe Control And Communications	Environmental controller	ready	5	sensor_bus; actuator_bus; lighting_control	environmental_controller	none	Submodule has component candidates and no allocation blocker.
Recipe Control And Communications	Recipe management	ready	4	operator_network; lighting_control	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Recipe Control And Communications	Remote monitoring	ready	4	operator_network; video_link	none	none	Submodule has component candidates and no allocation blocker.
Wet-Zone Power Distribution	Mains isolation panel	ready	5	mains_supply; protective_earth; service_accesses	rcd_isolator_panel	none	Submodule has component candidates and no allocation blocker.
Wet-Zone Power Distribution	Lighting power distribution	ready	4	mains_supply; lighting_control	none	none	Submodule has component candidates and no allocation blocker.
Wet-Zone Power Distribution	Low-voltage control power	ready	4	actuator_bus; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Food-Safe Safety And Protection	Leak containment	ready	4	washdown_boundary; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Food-Safe Safety And Protection	Operator interlocks	ready	4	service_accesses; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Food-Safe Safety And Protection	Food contact controls	ready	4	washdown_boundary; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Harvest Cleaning And Serviceability	Harvest access	ready	4	service_accesses; tray_support	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Harvest Cleaning And Serviceability	Cleaning access	ready	4	washdown_boundary; service_access	none	none	Submodule has component candidates and no allocation blocker.
Harvest Cleaning And Serviceability	Consumable service	ready	4	service_access; nutrient_loop	none	none	Submodule has component candidates and no allocation blocker.
Operator HMI And Labelling	Local HMI	ready	4	operator_network; service_access	none	none	Submodule has component candidates and no allocation blocker.
Operator HMI And Labelling	Labelling and workflows	ready	4	service_access	none	none	Submodule has component candidates and no allocation blocker.

Submodule Engineering Gate

submodule_engineering_review_required

0/28 submodules pass; 28 review, 0 blocked. Component words 122; linked requirements 9; carrier contracts 11/11; local-only interfaces 52. Review questions accepted 0/67; verification accepted 0/50; critical unpriced lines 8.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Growing Structure And Containment	Growing rack stack	review	5	2	3	0/0	0/2	0/3	1	0.6667	Review 2 local-only interface declaration(s): tray_support, service_access. 2 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
	Crop tray carriers	review	5	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): tray_support, nutrient_loop. 2 engineering review question(s) await acceptance. 1 verification

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											ion activity row(s) await accepted evidence.
Growing Structure And Containment	Enclosure panels	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): washdown_boundary, airflow. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Light Climate And Gas Environment	Horticultural lighting	review	5	3	3	0/0	0/2	0/3	1	0.6667	Review 3 local-only interface declaration(s): lighting_control, mains_supply, service_access. 2 engineering review question(s) await acceptance. 2 verification

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Light Climate And Gas Environment	Canopy air mixing	review	4	2	0	1/1	0/3	0/3	1	0.6667	Review 1 local-only interface declaration(s): airflow. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Light Climate And Gas	Humidity and tempera	review	5	3	0	2/2	0/4	0/3	0	0.6667	Review 1 local-only

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Environment	ture control										interface declaration(s): airflow. 4 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence.
Light Climate And Gas Environment	CO2 enrichment	review	5	3	0	2/2	0/4	0/4	1	0.6667	Review 1 local-only interface declaration(s): airflow. 4 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Nutrient And Water Transport Process	Reservoir and plumbing	review	5	2	0	0/0	0/2	0/2	1	0.6667	Review 2 local-only interface declaration(s): nutrient_loop, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Nutrient And Water Transport Process	Fertigation pumping	review	5	3	0	2/2	0/4	0/4	1	0.6667	Review 1 local-only interface declaration(s): nutrient_loop. 4 engineering review question(s) await acceptance. 3 verification activity

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Nutrient And Water Transport Process	Nutrient dosing	review	5	3	0	2/2	0/4	0/3	0	0.6667	Review 1 local-only interface declaration(s): nutrient_loop. 4 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence.
Nutrient And Water Transport Process	Filtration and sanitation	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): nutrient_loop, service_access. 2 engineering review question(s)

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											await acceptance. 1 verification activity row(s) await accepted evidence.
Crop And Process Instrumentation	Climate sensors	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): sensor_bus, airflow. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Crop And Process Instrumentation	Nutrient sensors	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): sensor_bus, nutrient_loop. 2 engineering review question(s) await acceptance. 1 verification activity

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											row(s) await acceptance. evidence.
Crop And Process Instrumentation	Crop observation	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): video_link, sensor_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await acceptance. evidence.
Recipe Control And Communications	Environmental controller	review	5	3	0	2/2	0/4	0/4	1	0.6667	Review 2 local-only interface declaration(s): sensor_bus, lighting_control. 4 engineering review question(s) await acceptance. 3 verification activity row(s) await acceptance. evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											e. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Recipe Control And Communications	Recipe management	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): operator_network, lighting_control. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Recipe Control And Communications	Remote monitoring	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): operator_network, video_link. 2 engineering review question(s) await

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											acceptance. 1 verification activity row(s) await accepted evidence.
Wet-Zone Power Distribution	Mains isolation panel	review	5	3	0	0/0	0/2	0/2	1	0.6667	Review 3 local-only interface declaration(s): mains_supply, protective_earth, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Wet-Zone Power Distribution	Lighting power distribution	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											ion(s): mains_supply, lighting_control. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Wet-Zone Power Distribution	Low-voltage control power	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): actuator_bus, sensor_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Food-Safe Safety And Protection	Leak containment	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): washdown_boundary, alarm_b

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											us. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Food-Safe Safety And Protection	Operator interlocks	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): service_access, alarm_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Food-Safe Safety And Protection	Food contact controls	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): washdown_boundary, service_access. 2 engineering review

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Harvest Cleaning And Serviceability	Harvest access	review	4	2	3	0/0	0/2	0/2	0	0.6667	Review 2 local-only interface declaration(s): service_access, tray_support. 2 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Harvest Cleaning And Serviceability	Cleaning access	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): washdown_boundary, service_access. 2 engineering review question(s) await acceptance

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											nce. 1 verification activity row(s) await accepted evidence.
Harvest Cleaning And Serviceability	Consumable service	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): service_access, nutrient_loop. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Operator HMI And Labelling	Local HMI	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): operator_network, service_access. 2 engineering review question(s) await acceptance. 1 verification activity

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
row(s) await accepted evidence.											
Operator HMI And Labelling	Labelling and workflows	review	4	1	0	0/0	0/2	0/1	0	0.6667	Review 1 local-only interface declaration(s): service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.

Module Engineering Gate

module_engineering_review_required

0/9 modules pass; 9 review, 0 blocked. Submodules/components: 28/122. Interface carriers 6/6; review questions accepted 0/66; critical unpriced lines 8 across 5 module(s).

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Growing Structure And Containme nt	review	3	14	3	0/0	0/10	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 10 engineerin g review question(s) await acceptanc e. 4 assumptio n row(s) need review or source evidence.
Light Climate And Gas Environme nt	review	4	19	3	2/2	0/15	3	0.86	3 critical BoM line(s) need source- backed cost evidence. 15 engineerin g review question(s) await acceptanc e. 9 assumptio n row(s) need review or source evidence.
Nutrient And Water Transport Process	review	4	19	0	2/2	0/9	2	0.71	2 critical BoM line(s) need source- backed cost evidence. 9 engineerin g review question(s) await acceptanc e. 5 assumptio n row(s) need review or

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
									source evidence.
Crop And Process Instrument ation	review	3	12	0	0/0	0/4	0	0.71	4 engineerin g review question(s) await acceptanc e. 1 assumptio n row(s) need review or source evidence.
Recipe Control And Communic ations	review	3	13	0	2/2	0/7	1	0.71	1 critical BoM line(s) need source- backed cost evidence. 7 engineerin g review question(s) await acceptanc e. 4 assumptio n row(s) need review or source evidence.
Wet-Zone Power Distributio n	review	3	13	0	0/0	0/5	1	0.71	1 critical BoM line(s) need source- backed cost evidence. 5 engineerin g review question(s) await acceptanc e. 3 assumptio n row(s) need review or source evidence.

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Food-Safe Safety And Protection	review	3	12	0	0/0	0/4	0	0.71	4 engineerin g review question(s) await acceptanc e.
Harvest Cleaning And Serviceabi lity	review	3	12	3	0/0	0/9	0	0.86	9 engineerin g review question(s) await acceptanc e. 3 assumptio n row(s) need review or source evidence.
Operator HMI And Labelling	review	2	8	0	0/0	0/3	0	0.71	3 engineerin g review question(s) await acceptanc e.

Module Review Roll-Up

4 ready, 5 sourcing-blocked, 0 needing attention.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Growing Structure And Containment	sourcing	3	14	0/0	envelope_length_m; envelope_width_m; footprint_m2	growing_rack_structure	1 critical line(s) need source-backed evidence before BoM pass.
Light Climate And Gas Environment	sourcing	4	19	2/2	envelope_length_m; envelope_width_m; footprint_m2	led_grow_lights; air_circulation_fans; co2_sensor_or_dosing	3 critical line(s) need source-backed evidence before BoM pass.
Nutrient And Water Transport Process	sourcing	4	19	2/2	none	reservoir_and_plumbing_set; nutrient_pump	2 critical line(s) need source-backed evidence before BoM pass.
Crop And Process Instrumentation	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.
Recipe Control And Communications	sourcing	3	13	2/2	none	environmental_controller	1 critical line(s) need source-backed evidence before BoM pass.
Wet-Zone Power Distribution	sourcing	3	13	0/0	none	rcd_isolator_panel	1 critical line(s) need source-backed evidence before BoM pass.
Food-Safe Safety And Protection	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.
Harvest Cleaning And Serviceability	ready	3	12	0/0	envelope_length_m; envelope_width_m; footprint_m2	none	No critical blockers; 12 candidate line(s) remain for later sourcing.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Operator HMI And Labelling	ready	2	8	0/0	none	none	No critical blockers; 8 candidate line(s) remain for later sourcing.

Engineering Review Pack

61 reviewer questions: 0 accepted, 41 ready, 20 needing review, 0 blocked. Coverage: 9 modules, 28 submodules, 3 interfaces, 2 calculations, 19 assumptions.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
assumption_resolution	major	needs review	CE-UKCA	Can the CE-UKCA assumption be accepted for architecture review, or must it be replaced with evidence?	Technical file, electrical risk assessment and declarations.	none	none	none	none
assumption_resolution	major	needs review	WRAS	Can the WRAS assumption be accepted for architecture review, or must it be replaced with evidence?	Approved fittings or evidence for water-contact components.	none	none	none	none
assumption_resolution	major	needs review	Environmental controller	Can the Environmental controller assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	control_compute_communication	none	none	none
assumption_resolution	major	needs review	Air circulation fans	Can the Air circulation fans assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	environmental_interface	none	none	none
assumption_resolution	major	needs review	CO2 sensor or dosing	Can the CO2 sensor or dosing	Admit supplier, manufacturer, MPN	environmental_interface	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				assumptio n be accepted for architecture review, or must it be replaced with evidence?	where applicable, unit cost, lead time and evidence reference through sourcing intake.				
assumptio n_resoluti on	major	needs review	LED grow lights	Can the LED grow lights assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	environme ntal_interf ace	none	none	none
assumptio n_resoluti on	major	needs review	Nutrient pump	Can the Nutrient pump assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	mass_fluid _transport _process	none	none	none
assumptio n_resoluti on	major	needs review	Reservoir and plumbing set	Can the Reservoir and plumbing set assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	mass_fluid _transport _process	none	none	none
assumptio n_resoluti on	major	needs review	RCD isolator panel	Can the RCD isolator panel assumption be	Admit supplier, manufacturer, MPN where applicable,	power_dis tribution	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				accepted for architecture review, or must it be replaced with evidence?	unit cost, lead time and evidence reference through sourcing intake.				
assumption_resolution	major	needs review	Growing rack structure	Can the Growing rack structure assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	structure_containment	none	none	none
assumption_resolution	major	needs review	control_compute_communication -> environmental_interface	Can the control_compute_communication -> environmental_interface assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	actuator_buses	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
assumption_resolution	major	needs review	control_compute_communication -> mass_fluid_transport_process	Can the control_compute_communication -> mass_fluid_transport_process assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	actuator_buses	none	none
assumption_resolution	major	needs review	environmental_interface -> mass_fluid_transport_process	Can the environmental_interface -> mass_fluid_transport_process assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	sensor_buses	none	none
assumption_resolution	major	needs review	Estimated CAPEX	Can the Estimated CAPEX assumption be accepted for architecture review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumption_resolution	major	needs review	Annual leafy-green yield	Can the Annual leafy-green yield assumption be accepted for architecture	Review the formula, input values and operating envelope before treating this metric	none	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				re review, or must it be replaced with evidence?	as a performance claim.				
assumption_resolution	major	needs review	Estimated annual OPEX	Can the Estimated annual OPEX assumption be accepted for architecture review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumption_resolution	major	needs review	Grow Unit Footprint	Can the Grow Unit Footprint assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	structure_containment; maintenance_serviceability; environmental_interface	none	none	none
assumption_resolution	major	needs review	Climate/Nutrient/Closure	Can the Climate/Nutrient/Closure assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	structure_containment; maintenance_serviceability; environmental_interface	none	none	none
assumption_resolution	major	needs review	Wet/Electrical Separation	Can the Wet/Electrical Separation assumption be accepted for architecture review,	Supply an engineering calculation , reviewer note, test plan or datasheet -backed rationale	structure_containment; maintenance_serviceability; environmental_interface	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				or must it be replaced with evidence?	for this envelope.				
calculation_envelope	major	needs review	Indicative service-clearance footprint	Does the Indicative service-clearance footprint calculation support the architecture envelope, or does the concept need revision?	Review maintenance access, nutrient tank access, airflow paths and electrical service clearances.	structure_containment; maintenance_service_ability; environmental_interface	none	farm_service_clearance_area_m2	none
calculation_envelope	major	ready	Vertical farm footprint check	Does the Vertical farm footprint check calculation support the architecture envelope, or does the concept need revision?	Confirm whether the envelope includes aisle, service clearance, reservoir and electrical cabinet space.	structure_containment; maintenance_service_ability; environmental_interface	none	farm_footprint_check_m2	none
interface_contract	major	ready	Recipe Control And Communications -> Light Climate And Gas Environment via actuator_bus	Is the actuator_bus interface between Recipe Control And Communications and Light Climate And Gas Environment correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	control_compute_communication; environmental_interface	actuator_bus	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Recipe Control And Communic	Is the actuator_bus interface	Interface-review evidence naming	control_compute_communicati	actuator_bus	none	Both endpoints expose the

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
			ations -> Nutrient And Water Transport Process via actuator_bus	between Recipe Control And Communications and Nutrient And Water Transport Process correctly owned and physically carried?	both endpoint owners, carrier submodules, capacity limits and any deferred checks.	mass_fluid_transport_process			required interface through at least one submodule carrier.
interface_contract	major	ready	Light Climate And Gas Environment -> Nutrient And Water Transport Process via sensor_buses	Is the sensor_buses interface between Light Climate And Gas Environment and Nutrient And Water Transport Process correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	environmental_interface; mass_fluid_transport_process	sensor_buses	none	Both endpoints expose the required interface through at least one submodule carrier.
module_allocation	major	ready	Recipe Control And Communications	Does Recipe Control And Communications have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	control_compute_communication	sensor_buses; actuator_bus; lighting_control; video_link; operator_network	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
module_all ocation	major	ready	Light Climate And Gas Environment	Does Light Climate And Gas Environment have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	environme ntal_interf ace	lighting_c ontrol; airflow; sensor_bu s; actuator_b us; mains_sup ply; service_ac cess	farm_foot print; farm_wet_ electrical_ separation ; farm_proc ess_closur e	none
module_all ocation	major	ready	Operator HMI And Labelling	Does Operator HMI And Labelling have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	hmi_ergon omics	operator_ network; service_ac cess; alarm_bus	none	none
module_all ocation	major	ready	Harvest Cleaning And Serviceability	Does Harvest Cleaning And Serviceability have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	maintenan ce_service ability	service_ac cess; washdown _boundary ; nutrient_lo op; tray_supp ort	farm_foot print; farm_wet_ electrical_ separation ; farm_proc ess_closur e	none
module_all ocation	major	ready	Nutrient And Water Transport Process	Does Nutrient And Water Transport Process have the right responsibility boundary, submodule split and interface ownership	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	mass_fluid _transport _process	nutrient_lo op; sensor_bu s; actuator_b us; service_ac cess; washdown _boundary	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for this concept?					
module_all ocation	major	ready	Wet-Zone Power Distribution	Does Wet-Zone Power Distribution have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	power_distribution	mains_supply; lighting_control; actuator_bus; protective_earth; service_access; sensor_buses	none	none
module_all ocation	major	ready	Food-Safe Safety And Protection	Does Food-Safe Safety And Protection have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	safety_protection	mains_supply; washdown_boundary; service_access; alarm_bus	none	none
module_all ocation	major	ready	Crop And Process Instrumentation	Does Crop And Process Instrumentation have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	sensing_instrumentation	sensor_buses; video_link; nutrient_loop; airflow	none	none
module_all ocation	major	ready	Growing Structure And Containment	Does Growing Structure And Containment have the right responsibility boundary, submodule split and interface	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	structure_containment	service_access; tray_support; washdown_boundary; mains_supply; nutrient_loop; airflow	farm_footprint; farm_wet_electrical_separation; farm_process_closure	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				ownership for this concept?					
submodul e_allocation	major	ready	Recipe Control And Communications / Environmental controller	Do the componen t candidate s and interfaces assigned to Environmental controller make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	control_co mpute_co mmunicati on	sensor_bu s; actuator_b us; lighting_c ontrol	none	none
submodul e_allocation	major	ready	Recipe Control And Communic ations / Recipe managem ent	Do the componen t candidate s and interfaces assigned to Recipe managem ent make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	control_co mpute_co mmunicati on	operator_ network; lighting_c ontrol	none	none
submodul e_allocation	major	ready	Recipe Control And Communic ations / Remote monitoring	Do the componen t candidate s and interfaces assigned to Remote monitoring make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	control_co mpute_co mmunicati on	operator_ network; video_link	none	none
submodul e_allocation	major	ready	Light Climate And Gas Environme nt / Canopy air mixing	Do the componen t candidate s and interfaces assigned to Canopy air mixing make engineerin	Reviewer note confirming the submodul e role, componen t allocation and any missing	environme ntal_interf ace	airflow; actuator_b us	farm_foot print; farm_wet_ electrical_ separation ; farm_proc ess_closur e	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	ready	Light Climate And Gas Environment / CO2 enrichment	g sense for its stated purpose?	carriers or capacity checks.	environmental_interface	airflow; sensor_buss; actuator_buss	farm_footprint; farm_wet_electrical_separation; farm_process_closure	none
				Do the component candidate s and interfaces assigned to CO2 enrichment make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Horticultural lighting make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Humidity and temperature control make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.				
submodule_allocation	major	ready	Operator HMI And Labelling / Labelling and workflows	Do the component candidate s and interfaces assigned to	Reviewer note confirming the submodule role, component	hmi_ergonomics	service_access	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
submodule_allocation	major	ready	Operator HMI And Labelling / Local HMI	Labelling and workflows make engineering sense for its stated purpose?	allocation and any missing carriers or capacity checks.				
				Do the component candidates and interfaces assigned to Local HMI make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	hmi_ergonomics	operator_network; service_access	none	none
submodule_allocation	major	ready	Harvest Cleaning And Serviceability / Cleaning access	Do the component candidates and interfaces assigned to Cleaning access make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	maintenance_serviceability	washdown_boundary; service_access	farm_footprint; farm_wet_electrical_separation; farm_process_closure	none
submodule_allocation	major	ready	Harvest Cleaning And Serviceability / Consumable service	Do the component candidates and interfaces assigned to Consumable service make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	maintenance_serviceability	service_access; nutrient_loop	farm_footprint; farm_wet_electrical_separation; farm_process_closure	none
submodule_allocation	major	ready	Harvest Cleaning And Serviceability /	Do the component candidates and interfaces	Reviewer note confirming the submodule role,	maintenance_serviceability	service_access; tray_support	farm_footprint; farm_wet_electrical_separation;	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
			Harvest access	assigned to Harvest access make engineering sense for its stated purpose?	component allocation and any missing carriers or capacity checks.			farm_process_closure	
submodule_allocation	major	ready	Nutrient And Water Transport Process / Fertigation pumping	Do the component candidate s and interfaces assigned to Fertigation pumping make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	mass_fluid_transport_process	nutrient_loop; actuator_buses; sensor_buses	none	none
submodule_allocation	major	ready	Nutrient And Water Transport Process / Filtration and sanitation	Do the component candidate s and interfaces assigned to Filtration and sanitation make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	mass_fluid_transport_process	nutrient_loop; service_access	none	none
submodule_allocation	major	ready	Nutrient And Water Transport Process / Nutrient dosing	Do the component candidate s and interfaces assigned to Nutrient dosing make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	mass_fluid_transport_process	nutrient_loop; actuator_buses; sensor_buses	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculations	Blockers
submodule_allocation	major	ready	Nutrient And Water Transport Process / Reservoir and plumbing	Do the component candidates and interfaces assigned to Reservoir and plumbing make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	mass_fluid_transport_process	nutrient_loop; service_access	none	none
submodule_allocation	major	ready	Wet-Zone Power Distribution / Lighting power distribution	Do the component candidates and interfaces assigned to Lighting power distribution make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	mains_supply; lighting_control	none	none
submodule_allocation	major	ready	Wet-Zone Power Distribution / Low-voltage control power	Do the component candidates and interfaces assigned to Low-voltage control power make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	actuator_buses; sensor_buses	none	none
submodule_allocation	major	ready	Wet-Zone Power Distribution / Mains isolation panel	Do the component candidates and interfaces assigned to Mains isolation panel make	Reviewer note confirming the submodule role, component allocation and any missing	power_distribution	mains_supply; protective_earth; service_access	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
submodule_allocation	major	ready	Food-Safe Safety And Protection / Food contact controls	engineering sense for its stated purpose?	carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Food contact controls make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	safety_protection	washdown_boundary ; service_access	none	none
submodule_allocation	major	ready	Food-Safe Safety And Protection / Leak containment	Do the component candidate s and interfaces assigned to Leak containment make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	safety_protection	washdown_boundary ; alarm_bus	none	none
submodule_allocation	major	ready	Food-Safe Safety And Protection / Operator interlocks	Do the component candidate s and interfaces assigned to Operator interlocks make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	safety_protection	service_access; alarm_bus	none	none
submodule_allocation	major	ready	Crop And Process Instrumentation / Climate sensors	Do the component candidate s and interfaces assigned to Climate sensors	Reviewer note confirming the submodule role, component allocation	sensing_instrumentation	sensor_buses; airflow	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				make engineering sense for its stated purpose?	and any missing carriers or capacity checks.				
submodule_allocation	major	ready	Crop And Process Instrumentation / Crop observation	Do the component candidate s and interfaces assigned to Crop observation make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	sensing_instrumentation	video_link; sensor_buses	none	none
submodule_allocation	major	ready	Crop And Process Instrumentation / Nutrient sensors	Do the component candidate s and interfaces assigned to Nutrient sensors make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	sensing_instrumentation	sensor_buses; nutrient_loop	none	none
submodule_allocation	major	ready	Growing Structure And Containment / Crop tray carriers	Do the component candidate s and interfaces assigned to Crop tray carriers make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	structure_containment	tray_support; nutrient_loop	farm_footprint; farm_wet_electrical_separation ; farm_process_closure	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
submodule_allocation	major	ready	Growing Structure And Containment / Enclosure panels	Do the component candidates and interfaces assigned to Enclosure panels make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	structure_containment	washdown_boundary; airflow	farm_footprint; farm_wet_electrical_separation; farm_process_closure	none

submodule_allocation	major	ready	Growing Structure And Containment / Growing rack stack	Do the component candidates and interfaces assigned to Growing rack stack make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	structure_containment	tray_support; service_access	farm_footprint; farm_wet_electrical_separation; farm_process_closure	none
----------------------	-------	-------	--	--	--	-----------------------	------------------------------	--	------

Engineering Sanity Checks

Check	Status	Value	Expected	Interpretation
Grow Unit Footprint	pass	3.36 m2	1 to 20 m2 for compact prototype grow units	3.36 m2 is a compact but workable footprint if stacked trays and service access are resolved.
Wet/Electrical Separation	pass	present	washdown boundary, mains supply and protective-earth interfaces present	The architecture declares a wet-zone boundary and protected electrical distribution before sourcing components.
Climate/Nutrient/Control Closure	pass	present	environmental, nutrient transport, sensing and control modules all present	Crop growth has a closed loop for light/climate, nutrient movement, sensing and controller action.

Engineering Calculation Ledger

2 calculations: 1 within deterministic envelope, 1 needing engineering review, 0 outside envelope, 0 blocked by missing inputs. These are arithmetic gates, not proof of real-world performance.

Calculation	Status	Formula	Result	Envelope	Interpretation	Evidence Required
Vertical farm footprint check	within envelope	envelope_length_m * envelope_width_m	3.36 m2	Calculated footprint should match parsed footprint within 1%.	3.36 m2 calculated from dimensions versus 3.36 m2 parsed footprint.	Confirm whether the envelope includes aisle, service clearance, reservoir and electrical cabinet space.
Indicative service-clearance footprint	needs review	footprint_m2 * 1.25	4.2 m2	Planning allowance only; layout needs crop tower count and operator-access review.	4.2 m2 allows a provisional 25% service envelope around the parsed footprint.	Review maintenance access, nutrient tank access, airflow paths and electrical service clearances.

Engineering Assumption Ledger

22 assumptions: 3 brief-supported, 3 model-present, 6 needing engineering review, 10 needing source evidence, 0 blocked. Architecture blockers 0; BoM blockers 10.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Envelope length	brief_requirement	brief supported	no	no	Envelope length = 2.4 m is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Envelope width	brief_requirement	brief supported	no	no	Envelope width = 1.4 m is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Footprint	brief_requirement	brief supported	no	no	Footprint = 3.36 m2 is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Annual leafy-green yield	derived_metric	review required	no	no	Annual leafy-green yield uses class_pack_default with low confidence.	Review the formula, input values and operating envelope before treating this metric as a performance claim.
Estimated CAPEX	derived_metric	source required	no	yes	Estimated CAPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Estimated annual OPEX	derived_metric	source required	no	yes	Estimated annual OPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Grow Unit Footprint	sanity_envelope	review required	no	no	Grow Unit Footprint currently evaluates as pass against 1 to 20 m2 for compact prototype grow units.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Wet/Electrical Separation	sanity_envelope	review required	no	no	Wet/Electrical Separation currently	Supply an engineering calculation,

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					evaluates as pass against washdown boundary, mains supply and protective-earth interfaces present.	reviewer note, test plan or datasheet-backed rationale for this envelope.
Climate/Nutrient/Control Closure	sanity_envelope	review required	no	no	Climate/Nutrient/Control Closure currently evaluates as pass against environmental, nutrient transport, sensing and control modules all present.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
control_compute_communication -> environmental_interface	interface_closure	model present	no	no	actuator_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
control_compute_communication -> mass_fluid_transport_process	interface_closure	model present	no	no	actuator_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
environmental_interface -> mass_fluid_transport_process	interface_closure	model present	no	no	sensor_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
Growing rack structure	critical_component	source required	no	yes	Growing rack structure is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
LED grow lights	critical_component	source required	no	yes	LED grow lights is a critical architecture-	Admit supplier, manufacturer, MPN where

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					derived component candidate with quantity 1 candidate.	applicable, unit cost, lead time and evidence reference through sourcing intake.
Air circulation fans	critical_component	source required	no	yes	Air circulation fans is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
CO2 sensor or dosing	critical_component	source required	no	yes	CO2 sensor or dosing is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Reservoir and plumbing set	critical_component	source required	no	yes	Reservoir and plumbing set is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Nutrient pump	critical_component	source required	no	yes	Nutrient pump is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Environmental controller	critical_component	source required	no	yes	Environmental controller is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
RCD isolator panel	critical_component	source required	no	yes	RCD isolator panel is a critical architecture-derived component candidate with	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					quantity 1 candidate.	through sourcing intake.
CE-UKCA	compliance_evidence	review required	no	no	Electrical safety and UKCA/CE file is relevant to this product class in UK/EU.	Technical file, electrical risk assessment and declarations.
WRAS	compliance_evidence	review required	no	no	Water fittings compliance is relevant to this product class in UK.	Approved fittings or evidence for water-contact components.

Engineering Verification Plan

24 activities: 15 ready for review, 4 open, 5 blocked.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Growing Structure And Containment architecture allocation	ready	design_review	Growing Structure And Containment	envelope_length_m; envelope_width_m; footprint_m2	service_access; tray_support; washdown_boundary; mains_supply; nutrient_loop; airflow	growing_rack_structure; aluminium_extrusion_upright; cross_brace_set; adjustable_leveling_foot; tray_slide_rail; food_safe_grow_tray; nft_channel_insert; seedling_raft; plus 6 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Growing Structure And Containment requirement calculations	ready	calculation	Growing Structure And Containment	envelope_length_m; envelope_width_m; footprint_m2	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Growing Structure And Containment critical components	blocked	source_evidence	Growing Structure And Containment	envelope_length_m; envelope_width_m; footprint_m2	none	growing_rack_structure	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved	growing_rack_structure

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	
Review Light Climate And Gas Environment architecture allocation	ready	design_review	Light Climate And Gas Environment	envelope_length_m; envelope_width_m; footprint_m2	lighting_control; airflow; sensor_bus; actuator_buses; mains_supply; service_access	led_grow_lights; dimmable_led_driver; lighting_suspension_bracket; ppfd_calibration_target; glare_shield; air_circulation_fans; fan_guard_grille; airflow_baffle_plate; plus 11 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Light Climate And Gas Environment requirement calculations	ready	calculation	Light Climate And Gas Environment	envelope_length_m; envelope_width_m; footprint_m2	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Collect source-backed evidence for Light Climate And Gas Environment critical components	blocked	source_evidence	Light Climate And Gas Environment	envelope_length_m; envelope_width_m; footprint_m2	none	led_grow_lights; air_circulation_fans; co2_sensor_or_dosing	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	led_grow_lights air_circulation_fans co2_sensor_or_dosing
Review Nutrient And Water Transport Process architecture allocation	ready	design_review	Nutrient And Water Transport Process	none	nutrient_loop; sensor_bus; actuator_buses; service_access; washdown_boundary	reservoir_and_plumbing_set; opaque_nutrient_reservoir; bulkhead_fitting_set; pvc_u_manifold; return_drain_header; nutrient_pump; pump_isolation_valve; check_valve; plus 11 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Collect source-backed evidence for Nutrient And Water Transport	blocked	source_evidence	Nutrient And Water Transport Process	none	none	reservoir_and_plumbing_set; nutrient_pump	Supplier name, manufacturer or equivalent, MPN where applicable,	reservoir_and_plumbing_set nutrient_pump

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Process critical components							unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	
Review Crop And Process Instrumentation architecture allocation	ready	design_review	Crop And Process Instrumentation	none	sensor_bus; video_link; nutrient_loop; airflow	temperature_humidity_probe; par_light_sensor; leaf_temperature_sensor; differential_pressure_switch; ph_probe; ec_probe; reservoir_level_sensor; solution_temperature_probe; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Review Recipe Control And Communications architecture allocation	ready	design_review	Recipe Control And Communications	none	sensor_bus; actuator_buses; lighting_control; video_link; operator_network	environmental_controller; din_rail_plc; analog_input_module; relay_output_module; real_time_clock_module; recipe_database; set_points_scheduler;	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						alarm_historian; plus 5 more	where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Collect source-backed evidence for Recipe Control And Communications critical components	blocked	source_evidence	Recipe Control And Communications	none	none	environmental_controller	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	environmental_controller
Review Wet-Zone Power Distribution architecture allocation	ready	design_review	Wet-Zone Power Distribution	none	mains_supply; lighting_control; actuator_buses; protective_earth; service_access; sensor_bus	rcd_isolator_panel; lockable_main_isolator; rcbo_breaker_bank; protective_earth_bar; surge_protection_device; lighting_bus_trunk; ip65_cable_gland_set; driver_mounting_rail; plus 5 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							issue is attached to the module.	
Collect source-backed evidence for Wet-Zone Power Distribution critical components	blocked	source_evidence	Wet-Zone Power Distribution	none	none	rcd_isolator_panel	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	rcd_isolator_panel
Review Food-Safe Safety And Protection architecture allocation	ready	design_review	Food-Safe Safety And Protection	none	mains_supply; washdown_boundary; service_access; alarm_bus	leak_detection_rope; bunded_base_tray; overflow_standpipe; floor_drain_adapter; door_interlock_switch; lighting_inhibit_relay; chemical_storage_latch; emergency_stop_button; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Review Harvest Cleaning And	ready	design_review	Harvest Cleaning And	envelope_length_m; envelope_width_m;	service_access; washdown_boundary;	slide_out_tray_handle; shelf_stop_latch;	Every submodule has a stated purpose	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Serviceability architecture allocation			Serviceability	footprint_m2	nutrient_loop; tray_support	harvest_work_ledge; removable_seedling_cassette; quick_release_manifold_union; removable_sump_screen; washdown_hose_connection; drying_fan_mode_switch; plus 4 more	and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Verify Harvest Cleaning And Serviceability requirement calculations	ready	calculation	Harvest Cleaning And Serviceability	envelope_length_m; envelope_width_m; footprint_m2	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Operator HMI And Labelling architecture allocation	ready	design_review	Operator HMI And Labelling	none	operator_network; service_access; alarm_bus	touchscreen_hmi; recipe_selector_control; alarm_buzzer_beacon; usb_data_export_port; crop_lot_label_holder; chemical_hazard_label_set; cleaning_checklist_card; electrical_isolation_placard	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify actuator_buss interface contract	ready	interface_review	Recipe Control And Communications -> Light Climate And Gas Environment	none	actuator_buss	environmental_controller; din_rail_plc; analog_input_module; relay_output_module; real_time_clock_module; air_circulation_fans; fan_guard_grille; airflow_baffle_plate; plus 11 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify actuator_buss interface contract	ready	interface_review	Recipe Control And Communications -> Nutrient And Water Transport Process	none	actuator_buss	environmental_controller; din_rail_plc; analog_input_module; relay_output_module; real_time_clock_module; nutrient_pump; pump_isolat	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid,	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						ion_valve; check_valve; plus 7 more	mechanical or data capacity is validated by the next detailed design pass.	
Verify sensor_bus interface contract	ready	interface_review	Light Climate And Gas Environment -> Nutrient And Water Transport Process	none	sensor_bus	dehumidifier_module; heat_exchanger_coil; condensate_pump; supply_air_temperature_probe; humidity_control_relay; co2_sensor_or_dosing; ndir_co2_sensor_head; co2_solenoid_valve; plus 12 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Plan evidence for CE-UKCA	open	compliance_review	System Compliance	none	none	none	Technical file, electrical risk assessment and declarations .	Compliance evidence not yet collected in scratch prototype.
Plan evidence for WRAS	open	compliance_review	System Compliance	none	none	none	Approved fittings or evidence for water-contact components .	Compliance evidence not yet collected in scratch prototype.
Review risk control for Crop loss from pump failure	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any	Risk evidence not yet reviewed in scratch prototype.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							residual high-risk item has an owner before publication.	
Review risk control for Electrical fault in humid grow area	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.

Verification Evidence Ledger

0 accepted, 0 rejected, 0 deferred, 19 pending, 0 blocked without evidence. Review coverage 0; acceptance ratio 0. Source-evidence activities remain in sourcing intake.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review Growing Structure And Containment architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Growing Structure And Containment requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Growing Structure And Containment critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Light Climate And Gas Environment architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Light Climate And Gas Environment requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Light Climate And Gas Environment critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Nutrient And Water Transport Process architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Nutrient And Water Transport Process critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Crop And Process Instrumentation architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review Recipe Control And Communications architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Collect source-backed evidence for Recipe Control And Communications critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Wet-Zone Power Distribution architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Wet-Zone Power Distribution critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Food-Safe Safety And Protection architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review Harvest Cleaning And Serviceability architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Harvest Cleaning And Serviceability requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Review Operator HMI And Labelling architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify actuator_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify actuator_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify sensor_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Plan evidence for CE-UKCA	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for WRAS	pending	compliance_review			Collect reviewer evidence through the verification intake template.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review risk control for Crop loss from pump failure	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Electrical fault in humid grow area	pending	compliance_review			Collect reviewer evidence through the verification intake template.

Growing Structure And Containment

Carries stacked crop trays, lighting and service access within a compact 3.36 m2 footprint.

- service_access
- tray_support
- washdown_boundary
- mains_supply

Growing rack stack
Supports stacked trays, lighting bars and plant mass.

Components: Growing rack structure, aluminium extrusion upright, cross brace set, adjustable levelling foot, tray slide rail

Crop tray carriers
Holds channels, mats and seedlings at controlled spacing.

Components: food-safe grow tray, NFT channel insert, seedling raft, root-zone cover, tray drain fitting

Enclosure panels
Separates humid grow volume from room air and service zones.

Components: washdown side panel, clear inspection door, EPDM door gasket, condensate drip channel

Light Climate And Gas Environment

Controls photons, airflow, humidity and CO2 around the crop canopy.

- lighting_control
- airflow
- sensor_bus
- actuator_bus
- mains_supply

Horticultural lighting
Delivers crop-specific photosynthetic photon flux to each shelf.

Components: LED grow lights, dimmable LED driver, lighting suspension bracket, PPFD calibration target, glare shield

Canopy air mixing
Moves air across leaves to reduce stagnant humidity pockets.

Components: Air circulation fans, fan guard grille, airflow baffle plate, replaceable intake filter

Humidity and temperature control
Maintains crop vapor-pressure deficit and sensible heat balance.

Components: dehumidifier module, heat exchanger coil, condensate pump, supply air temperature probe, humidity control relay

CO2 enrichment
Measures and doses CO2 when the growing chamber is closed.

Components: CO2 sensor or dosing, NDIR CO2 sensor head, CO2 solenoid valve, gas regulator, CO2 distribution tube

Nutrient And Water Transport Process

Stores, doses, filters and circulates nutrient solution through crop trays.

- nutrient_loop
- sensor_bus
- actuator_bus
- service_access
- washdown_boundary

Reservoir and plumbing

Contains nutrient solution and routes it to shelves.

Components: Reservoir and plumbing set, opaque nutrient reservoir, bulkhead fitting set, PVC-U manifold, return drain header

Fertigation pumping

Moves nutrient solution at controlled flow rate.

Components: Nutrient pump, pump isolation valve, check valve, flow meter, pump vibration mount

Nutrient dosing

Adds nutrient concentrates and pH correction to the reservoir.

Components: peristaltic dosing pump, pH dosing pump, chemical suction lance, anti-siphon valve, mixing eductor

Filtration and sanitation

Keeps the nutrient loop clean and serviceable.

Components: inline mesh filter, UV steriliser, drain valve, sample port

Crop And Process Instrumentation

Measures climate, nutrient chemistry and crop state for control and review.

- sensor_bus
- video_link
- nutrient_loop
- airflow

Climate sensors

Measures canopy temperature, humidity and light level.

Components: temperature humidity probe, PAR light sensor, leaf temperature sensor, differential pressure switch

Nutrient sensors

Measures solution chemistry and reservoir level.

Components: pH probe, EC probe, reservoir level sensor, solution temperature probe

Crop observation

Captures visual crop evidence and growth state.

Components: RGB crop camera, camera light shield, time-lapse controller, calibration colour card

Recipe Control And Communications

Runs crop recipes, controls actuators and records operating history.

- sensor_bus
- actuator_bus
- lighting_control
- video_link
- operator_network

Environmental controller

Coordinates lighting, fertigation, climate and alarms.

Components: Environmental controller, DIN-rail PLC, analog input module, relay output module, real-time clock module

Recipe management

Stores crop recipes and set-point schedules.

Components: recipe database, set-point scheduler, alarm historian, data export module

Remote monitoring

Provides operator status and alerts outside the unit.

Components: industrial Ethernet switch, Wi-Fi gateway, MQTT telemetry bridge, SMS alarm modem

Wet-Zone Power Distribution

Distributes mains and low-voltage power to lighting, pumps, controls and safety devices.

- mains_supply
- lighting_control
- actuator_bus
- protective_earth

Mains isolation panel

Provides incoming isolation and branch protection.

Components: RCD isolator panel, lockable main isolator, RCBO breaker bank, protective earth bar, surge protection device

Lighting power distribution

Feeds and switches LED drivers by shelf.

Components: lighting bus trunk, IP65 cable gland set, driver mounting rail, fused spur module

Low-voltage control power

Feeds sensors, controller and communication devices.

Components: 24 V DC power supply, DC distribution terminal, UPS buffer module, control cabinet heater

Food-Safe Safety And Protection

Controls electrical, water, chemical and operator risks in a humid cultivation unit.

- mains_supply
- washdown_boundary
- service_access
- alarm_bus

Leak containment

Detects and contains nutrient leaks before they reach electrical zones.

Components: leak detection rope, bunded base tray, overflow standpipe, floor drain adapter

Operator interlocks

Reduces exposure to bright light, moving fans and chemicals during service.

Components: door interlock switch, lighting inhibit relay, chemical storage latch, emergency stop button

Food contact controls

Keeps crop-facing components cleanable and traceable.

Components: food-grade tubing set, cleaning validation swab kit, material traceability label, washdown SOP placard

Harvest Cleaning And Serviceability

Supports tray removal, cleaning, filter changes and routine crop operations.

- service_access
- washdown_boundary
- nutrient_loop

Harvest access

Allows ergonomic crop removal and replanting.

Components: slide-out tray handle, shelf stop latch, harvest work ledge, removable seedling cassette

Cleaning access

Allows sanitation of wet and crop-contact areas.

Components: quick-release manifold union, removable sump screen, washdown hose connection, drying fan mode switch

Consumable service

Makes filters, probes and dosing consumables replaceable.

Components: filter service hatch, probe calibration cup, dosing bottle tray, spare gasket kit

Operator HMI And Labelling

Shows grow state, alarms, recipes and safe operating instructions.

operator_network	service_access	alarm_bus
------------------	----------------	-----------

Local HMI

Provides local status, recipe selection and alarm acknowledgement.

Components: touchscreen HMI, recipe selector control, alarm buzzer beacon, USB data export port

Labelling and workflows

Communicates hazards, cleaning steps and crop lot identity.

Components: crop lot label holder, chemical hazard label set, cleaning checklist card, electrical isolation placard

Bill of Materials

Pricing status: architecture-derived candidate BoM only. Costs, suppliers, manufacturers and part numbers have not been independently sourced.

Total CAPEX BoM: **£0**

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
Growing rack structure	1	candidate	unpriced	unpriced				
aluminium extrusion upright	1	candidate	unpriced	unpriced				
cross brace set	1	candidate	unpriced	unpriced				
adjustable levelling foot	1	candidate	unpriced	unpriced				
tray slide rail	1	candidate	unpriced	unpriced				
food-safe grow tray	1	candidate	unpriced	unpriced				
NFT channel insert	1	candidate	unpriced	unpriced				
seedling raft	1	candidate	unpriced	unpriced				
root-zone cover	1	candidate	unpriced	unpriced				
tray drain fitting	1	candidate	unpriced	unpriced				
washdown side panel	1	candidate	unpriced	unpriced				
clear inspection door	1	candidate	unpriced	unpriced				
EPDM door gasket	1	candidate	unpriced	unpriced				
condensate drip channel	1	candidate	unpriced	unpriced				
LED grow lights	1	candidate	unpriced	unpriced				
dimnable LED driver	1	candidate	unpriced	unpriced				
lighting suspension bracket	1	candidate	unpriced	unpriced				
PPFD calibration target	1	candidate	unpriced	unpriced				
glare shield	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
Air circulation fans	1	candidate	unpriced	unpriced				
fan guard grille	1	candidate	unpriced	unpriced				
airflow baffle plate	1	candidate	unpriced	unpriced				
replaceable intake filter	1	candidate	unpriced	unpriced				
dehumidifier module	1	candidate	unpriced	unpriced				
heat exchanger coil	1	candidate	unpriced	unpriced				
condensate pump	1	candidate	unpriced	unpriced				
supply air temperature probe	1	candidate	unpriced	unpriced				
humidity control relay	1	candidate	unpriced	unpriced				
CO2 sensor or dosing	1	candidate	unpriced	unpriced				
NDIR CO2 sensor head	1	candidate	unpriced	unpriced				
CO2 solenoid valve	1	candidate	unpriced	unpriced				
gas regulator	1	candidate	unpriced	unpriced				
CO2 distribution tube	1	candidate	unpriced	unpriced				
Reservoir and plumbing set	1	candidate	unpriced	unpriced				
opaque nutrient reservoir	1	candidate	unpriced	unpriced				
bulkhead fitting set	1	candidate	unpriced	unpriced				
PVC-U manifold	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
return drain header	1	candidate	unpriced	unpriced				
Nutrient pump	1	candidate	unpriced	unpriced				
pump isolation valve	1	candidate	unpriced	unpriced				
check valve	1	candidate	unpriced	unpriced				
flow meter	1	candidate	unpriced	unpriced				
pump vibration mount	1	candidate	unpriced	unpriced				
peristaltic dosing pump	1	candidate	unpriced	unpriced				
pH dosing pump	1	candidate	unpriced	unpriced				
chemical suction lance	1	candidate	unpriced	unpriced				
anti-siphon valve	1	candidate	unpriced	unpriced				
mixing eductor	1	candidate	unpriced	unpriced				
inline mesh filter	1	candidate	unpriced	unpriced				
UV steriliser	1	candidate	unpriced	unpriced				
drain valve	1	candidate	unpriced	unpriced				
sample port	1	candidate	unpriced	unpriced				
temperature humidity probe	1	candidate	unpriced	unpriced				
PAR light sensor	1	candidate	unpriced	unpriced				
leaf temperature sensor	1	candidate	unpriced	unpriced				
differential pressure switch	1	candidate	unpriced	unpriced				
pH probe	1	candidate	unpriced	unpriced				
EC probe	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
reservoir level sensor	1	candidate	unpriced	unpriced				
solution temperature probe	1	candidate	unpriced	unpriced				
RGB crop camera	1	candidate	unpriced	unpriced				
camera light shield	1	candidate	unpriced	unpriced				
time-lapse controller	1	candidate	unpriced	unpriced				
calibration colour card	1	candidate	unpriced	unpriced				
Environmental controller	1	candidate	unpriced	unpriced				
DIN-rail PLC	1	candidate	unpriced	unpriced				
analog input module	1	candidate	unpriced	unpriced				
relay output module	1	candidate	unpriced	unpriced				
real-time clock module	1	candidate	unpriced	unpriced				
recipe database	1	candidate	unpriced	unpriced				
set-point scheduler	1	candidate	unpriced	unpriced				
alarm historian	1	candidate	unpriced	unpriced				
data export module	1	candidate	unpriced	unpriced				
industrial Ethernet switch	1	candidate	unpriced	unpriced				
Wi-Fi gateway	1	candidate	unpriced	unpriced				
MQTT telemetry bridge	1	candidate	unpriced	unpriced				
SMS alarm modem	1	candidate	unpriced	unpriced				
RCD isolator panel	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
lockable main isolator	1	candidate	unpriced	unpriced				
RCBO breaker bank	1	candidate	unpriced	unpriced				
protective earth bar	1	candidate	unpriced	unpriced				
surge protection device	1	candidate	unpriced	unpriced				
lighting bus trunk	1	candidate	unpriced	unpriced				
IP65 cable gland set	1	candidate	unpriced	unpriced				
driver mounting rail	1	candidate	unpriced	unpriced				
fused spur module	1	candidate	unpriced	unpriced				
24 V DC power supply	1	candidate	unpriced	unpriced				
DC distribution terminal	1	candidate	unpriced	unpriced				
UPS buffer module	1	candidate	unpriced	unpriced				
control cabinet heater	1	candidate	unpriced	unpriced				
leak detection rope	1	candidate	unpriced	unpriced				
bunded base tray	1	candidate	unpriced	unpriced				
overflow standpipe	1	candidate	unpriced	unpriced				
floor drain adapter	1	candidate	unpriced	unpriced				
door interlock switch	1	candidate	unpriced	unpriced				
lighting inhibit relay	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
chemical storage latch	1	candidate	unpriced	unpriced				
emergency stop button	1	candidate	unpriced	unpriced				
food-grade tubing set	1	candidate	unpriced	unpriced				
cleaning validation swab kit	1	candidate	unpriced	unpriced				
material traceability label	1	candidate	unpriced	unpriced				
washdown SOP placard	1	candidate	unpriced	unpriced				
slide-out tray handle	1	candidate	unpriced	unpriced				
shelf stop latch	1	candidate	unpriced	unpriced				
harvest work ledge	1	candidate	unpriced	unpriced				
removable seedling cassette	1	candidate	unpriced	unpriced				
quick-release manifold union	1	candidate	unpriced	unpriced				
removable sump screen	1	candidate	unpriced	unpriced				
washdown hose connection	1	candidate	unpriced	unpriced				
drying fan mode switch	1	candidate	unpriced	unpriced				
filter service hatch	1	candidate	unpriced	unpriced				
probe calibration cup	1	candidate	unpriced	unpriced				
dosing bottle tray	1	candidate	unpriced	unpriced				
spare gasket kit	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
touchscreen HMI	1	candidate	unpriced	unpriced				
recipe selector control	1	candidate	unpriced	unpriced				
alarm buzzer beacon	1	candidate	unpriced	unpriced				
USB data export port	1	candidate	unpriced	unpriced				
crop lot label holder	1	candidate	unpriced	unpriced				
chemical hazard label set	1	candidate	unpriced	unpriced				
cleaning checklist card	1	candidate	unpriced	unpriced				
electrical isolation placard	1	candidate	unpriced	unpriced				

Cost And Sourcing

Sourcing Admission

Status	Candidate Lines	Admitted Prices	Unpriced Lines	Unpriced Critical Lines
not_started	122	0	122	8

Critical Sourcing Worklist

Component	Qty	Reason
Growing rack structure	1 candidate	Required class-critical line must be sourced before BoM can pass.
LED grow lights	1 candidate	Required class-critical line must be sourced before BoM can pass.
Air circulation fans	1 candidate	Required class-critical line must be sourced before BoM can pass.
CO2 sensor or dosing	1 candidate	Required class-critical line must be sourced before BoM can pass.
Reservoir and plumbing set	1 candidate	Required class-critical line must be sourced before BoM can pass.
Nutrient pump	1 candidate	Required class-critical line must be sourced before BoM can pass.
Environmental controller	1 candidate	Required class-critical line must be sourced before BoM can pass.
RCD isolator panel	1 candidate	Required class-critical line must be sourced before BoM can pass.

No rejected source records.

BoM Sourcing Ledger

0/122 priced lines admitted. Critical coverage 0/8. Rejected evidence records: 0. Cost/manufacturer/MPN fields stay blank until evidence is admitted.

Component identity check: no duplicate componentWordId groups detected.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
Growing rack structure	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
aluminium extrusion upright	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
cross brace set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
adjustable levelling foot	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
tray slide rail	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
food-safe grow tray	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
NFT channel insert	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
seedling raft	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
root-zone cover	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
tray drain fitting	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
washdown side panel	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
clear inspection door	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
EPDM door gasket	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
condensate drip channel	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
LED grow lights	criticalunpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
dimmmable LED driver	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
lighting suspension bracket	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
PPFD calibration target	candidateunpriced	unique	candidate	unpriced					Source after critical lines or

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									when detailed procurement planning starts.
glare shield	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Air circulation fans	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
fan guard grille	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
airflow baffle plate	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
replaceable intake filter	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
dehumidifier module	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
heat exchanger coil	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
condensate pump	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
supply air temperature probe	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
humidity control relay	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
CO2 sensor or dosing	critical unpriced	unique	critical	unpriced					Collect admissible supplier/manufacture r/MPN/unit -cost evidence before

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									BoM can pass.
NDIR CO2 sensor head	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
CO2 solenoid valve	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
gas regulator	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
CO2 distribution tube	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Reservoir and plumbing set	criticalunpriced	unique	critical	unpriced					Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
opaque nutrient reservoir	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
bulkhead fitting set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
PVC-U manifold	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
return drain header	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Nutrient pump	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
pump isolation valve	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Showing first 40 of 122 ledger rows; export CSV for the full line-by-line ledger.

0 source-backed claims, 32 missing-source claims, 0 provenance violations. Critical missing-source claims: 32. Supplier, manufacturer, MPN and cost fields must appear here with source refs before procurement use.

Line	Claim Field	Status	Value	Source Ref	Next Action
Growing rack structure	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Growing rack structure	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Growing rack structure	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Growing rack structure	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
LED grow lights	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
LED grow lights	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
LED grow lights	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
LED grow lights	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Air circulation fans	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Air circulation fans	manufacturer	missing source			Admit source-backed

Line	Claim Field	Status	Value	Source Ref	Next Action
					supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Air circulation fans	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Air circulation fans	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
CO2 sensor or dosing	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
CO2 sensor or dosing	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
CO2 sensor or dosing	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
CO2 sensor or dosing	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Reservoir and plumbing set	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Reservoir and plumbing set	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Reservoir and plumbing set	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost

Line	Claim Field	Status	Value	Source Ref	Next Action
					evidence before BoM can pass.
Reservoir and plumbing set	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Nutrient pump	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Nutrient pump	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Nutrient pump	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Nutrient pump	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Environmental controller	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Environmental controller	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Environmental controller	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Environmental controller	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Line	Claim Field	Status	Value	Source Ref	Next Action
RCD isolator panel	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
RCD isolator panel	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
RCD isolator panel	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
RCD isolator panel	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Source Reference Quality Gate

No sourcing evidence references have been admitted. Manufacturer, MPN and cost fields must remain blank until source-backed evidence is admitted through sourcing intake.

BoM Evidence Trace Matrix

0/122 BoM row(s) are production-eligible; 0 protocol-only, 0 source-reference blocked, 8 critical unsourced. Priced-review rows 0; procurement use no; next row bom-trace:structure_containment_growing_rack_stack_growing_rack_structure.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
Growing rack structure	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
aluminium extrusion upright	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
cross brace set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
adjustable levelling foot	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
tray slide rail	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
food-safe grow tray	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
NFT channel insert	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
seedling raft	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
root-zone cover	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
tray drain fitting	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
washdown side panel	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
clear inspection door	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
EPDM door gasket	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
condensate drip channel	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
LED grow lights	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
dimmmable LED driver	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
lighting suspension bracket	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									critical sourcing is complete or procurement planning starts.
PPFD calibration target	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
glare shield	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Air circulation fans	critical unsource	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
fan guard grille	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
airflow baffle plate	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
replaceable intake filter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
dehumidifier module	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
heat exchanger coil	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
condensate pump	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
supply air temperature probe	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
humidity control relay	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
CO2 sensor or dosing	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
NDIR CO2 sensor head	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
CO2 solenoid valve	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
gas regulator	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
CO2 distribution tube	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
Reservoir and plumbing set	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
opaque nutrient reservoir	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
bulkhead fitting set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
PVC-U manifold	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
return drain header	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
Nutrient pump	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
pump isolation valve	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
check valve	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
flow meter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
pump vibration mount	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
peristaltic dosing pump	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
pH dosing pump	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
chemical suction lance	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
anti-siphon valve	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
mixing eductor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
inline mesh filter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
UV steriliser	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 122 BoM trace rows; export CSV for the full trace.

BoM Evidence Closure Plan

122 closure row(s): 8 ready, 0 blocked, 114 deferred. Actions: 8 collect source, 0 repair, 0 replace protocol, 114 candidate deferrals. Procurement blockers 8; priced-review blockers 8; next row bom-close:control_compute_communication_environmental_controller_environmental_controller.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
1	Environmental controller	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
2	Air circulation fans	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
3	CO2 sensor or dosing	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
4	LED grow lights	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
5	Nutrient pump	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
6	Reservoir and plumbing set	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
7	RCD isolator panel	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
8	Growing rack structure	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
9	analog input module	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
10	DIN-rail PLC	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
11	real-time clock module	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
12	relay output module	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is	no	no	Keep as candidate-only until critical sourcing is complete or

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
						promoted to procurement scope			procurement planning starts.
13	alarm historian	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
14	data export module	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
15	recipe database	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
16	set-point scheduler	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
17	industrial Ethernet switch	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to	no	no	Keep as candidate-only until critical sourcing is complete or procurement

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
						procurement scope			planning starts.
18	MQTT telemetry bridge	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
19	SMS alarm modem	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
20	Wi-Fi gateway	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
21	airflow baffle plate	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
22	fan guard grille	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
23	replaceable intake filter	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
24	CO2 distribution tube	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
25	CO2 solenoid valve	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
26	gas regulator	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
27	NDIR CO2 sensor head	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
28	dimnable LED driver	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
29	glare shield	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
30	lighting suspension bracket	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
31	PPFD calibration target	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
32	condensate pump	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
33	dehumidifier module	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
34	heat exchanger coil	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
35	humidity control relay	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
36	supply air temperature probe	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
37	chemical hazard label set	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
38	cleaning checklist card	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
39	crop lot label holder	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
40	electrical isolation placard	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
41	alarm buzzer beacon	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
42	recipe selector control	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
43	touchscreen HMI	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
44	USB data export port	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
45	drying fan mode switch	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
46	quick-release manifold union	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
47	removable sump screen	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
48	washdown hose connection	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
49	dosing bottle tray	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
50	filter service hatch	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 122 BoM closure rows; export CSV for the full queue.

Sourcing Batch Plan

2 batch(es): 1 active, 0 waiting, 1 deferred. Active rows 8; deferred rows 114. Critical source rows 8; repair rows 0; protocol replacements 0. Next batch sourcing-batch:2:critical_source_collection; next item critical_source_collection:1:environmental_controller.

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	Environmental controller	Environmental controller; Environmental controller manufacturer part number; Environmental controller catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Air circulation fans	Air circulation fans; Air circulation fans manufacturer part number; Air circulation fans catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	CO2 sensor or dosing	CO2 sensor or dosing; CO2 sensor or dosing manufacturer part number; CO2 sensor or dosing catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	LED grow lights	LED grow lights; LED grow lights manufacturer part number; LED grow lights catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Nutrient pump	Nutrient pump; Nutrient pump manufacturer part number; Nutrient pump catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical	active	blocker	Reservoir and	Reservoir and	supplierName;	yes	source evidence

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
	Source Evidence			plumbing set	plumbing set; Reservoir and plumbing set manufacturer part number; Reservoir and plumbing set catalogue price GBP	manufacturer; mpn; unitCostGbp; evidence.ref		admitted
2	Collect Critical Source Evidence	active	blocker	RCD isolator panel	RCD isolator panel; RCD isolator panel manufacturer part number; RCD isolator panel catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Growing rack structure	Growing rack structure; Growing rack structure manufacturer part number; Growing rack structure catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
6	Defer Candidate-Only Sourcing	deferred	minor	analog input module	analog input module; analog input module manufacturer part number; analog input module catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
6	Defer Candidate-Only Sourcing	deferred	minor	DIN-rail PLC	DIN-rail PLC; DIN-rail PLC manufacturer part number; DIN-rail PLC catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	real-time clock module	real-time clock module; real-time clock module manufacturer part number; real-time clock module catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	relay output module	relay output module; relay output module manufacturer part number; relay output module catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	alarm historian	alarm historian; alarm historian manufacturer part number; alarm historian catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	data export module	data export module; data export module manufacturer part number; data export module catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
6	Defer Candidate-Only Sourcing	deferred	minor	recipe database	recipe database; recipe database manufacturer part number; recipe database catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	set-point scheduler	set-point scheduler; set-point scheduler manufacturer part number; set-point scheduler catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	industrial Ethernet switch	industrial Ethernet switch; industrial Ethernet switch manufacturer part number; industrial Ethernet switch catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	MQTT telemetry bridge	MQTT telemetry bridge; MQTT telemetry bridge manufacturer part number; MQTT telemetry bridge catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
6	Defer Candidate-Only Sourcing	deferred	minor	SMS alarm modem	SMS alarm modem; SMS alarm modem manufacturer part number; SMS alarm modem catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	Wi-Fi gateway	Wi-Fi gateway; Wi-Fi gateway manufacturer part number; Wi-Fi gateway catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

Procurement Readiness Gate

procurement_not_started

0/7 procurement areas pass; 2 review, 5 blocked. Critical production-eligible BoM rows 0/8; procurement-blocking closure rows 8; active sourcing rows 8; deferred sourcing rows 114. Source quality no_sourcing_evidence; evidence authenticity no_evidence; costing costing_not_started; BoM admission candidate_bom_authorized. Can use for procurement: no. Next action: Collect production-quality source evidence for critical BoM rows before procurement use..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture review	review	0.33	architecture_freeze_review_required; structurally ready yes, independent review accepted no.	none	Collect accepted engineering/interface/requirement review evidence before procurement use.
bom evidence trace	blocked	0	0/8 critical rows production-eligible; 8 critical unsourced, 0 protocol-only, 0 source-reference blocked.	structure_containment_growing_rack_stack_growing_rack_structure: Collect admissible source evidence for this critical line before BoM can pass. environmental_interface_horticultural_lighting_led_grow_lights: Collect admissible source evidence for this critical line before BoM can pass. environmental_interface_canopy_air_mixing_air_circulation_fans: Collect admissible source evidence for this critical line before BoM can pass. environmental_interface_co2_enrichment_co2_sensor_or_dosing: Collect admissible source evidence for this critical line before BoM can pass. mass_fluid_transport_process_reservoir_plumbing_reservoir_and_plumbing_set: Collect admissible source evidence for this critical line before BoM can pass. plus 3 more	Collect production-quality source evidence for critical BoM rows before procurement use.
sourcing batch closure	blocked	0.9344	8 procurement-blocking closure row(s); 8 active sourcing row(s), 114 deferred row(s).	bom-close:control_compute_communication_environmental_controller_environmental_controller: Collect admissible source evidence for this critical line before BoM can pass. bom-close:environmental_interface_canopy_air_mixing_air_circulation_fans: Collect admissible source	Work the active sourcing batches until procurement-blocking closure rows are closed.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				evidence for this critical line before BoM can pass. bom-close:environmental_interface_co2_enrichment_co2_sensor_or_dosing: Collect admissible source evidence for this critical line before BoM can pass. bom-close:environmental_interface_horticultural_lighting_led_grow_lights: Collect admissible source evidence for this critical line before BoM can pass. bom-close:mass_fluid_transport_process_fermentation_pumping_nutrient_pump: Collect admissible source evidence for this critical line before BoM can pass. plus 3 more	
source reference quality	blocked	0	no_sourcing_evidence; 0/0 source-reference row(s) pass; 0 protocol, 0 placeholder.	none	Admit production source evidence before procurement readiness can start.
evidence authenticity	blocked	0	no_evidence; 0/0 evidence rows production-ready; 0 protocol.	No evidence records are present.	Add source and reviewer evidence before procurement readiness can start.
bom costing	blocked	0.14	costing_not_started ; critical priced 0/8; BoM total 0 GBP.	structure_containment_growing_rack_stack_growing_rack_structure: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. environmental_interface_horticultural_lighting_led_grow_lights: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. environmental_interface_canopy_air_mixing_air_circulation_fans: Collect admissible	Close costing blockers before procurement use.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. environmental_interface_co2_enrichment_co2_sensor_or_dosing: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. mass_fluid_transport_process_reservoir_plumbing_reservoir_and_plumbing_set: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 46 more	
prototype procurement policy	review	0	candidate_bom_authorized; display mode candidate_only; prototype procurement use allowed no.	none	Prototype policy keeps procurement use disabled until external procurement review and approval are explicitly added.

Sourcing Evidence Pack

Evidence packets request source data only. They do not admit supplier, manufacturer, MPN or cost claims.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Growing rack structure	1 candidate	Growing rack structure; Growing rack structure manufacturer part number; Growing rack structure catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
LED grow lights	1 candidate	LED grow lights; LED grow lights manufacturer part number; LED grow lights catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Air circulation fans	1 candidate	Air circulation fans; Air circulation fans manufacturer part number; Air circulation fans catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
CO2 sensor or dosing	1 candidate	CO2 sensor or dosing; CO2 sensor or dosing manufacturer part number; CO2 sensor or dosing catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Reservoir and plumbing set	1 candidate	Reservoir and plumbing set; Reservoir and plumbing set manufacturer part number; Reservoir and plumbing set catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Nutrient pump	1 candidate	Nutrient pump; Nutrient pump manufacturer part number; Nutrient pump catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Environmental controller	1 candidate	Environmental controller; Environmental controller manufacturer part number; Environmental controller catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
RCD isolator panel	1 candidate	RCD isolator panel; RCD isolator panel manufacturer part number; RCD isolator panel catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Cost model status: pending sourcing. Architecture is being reviewed before cost claims are admitted.

Cost Bucket	Estimate
CAPEX	£0
Annual OPEX	£0
NRE	£25,000

No admitted supplier spend yet. Supplier rows appear only after source-backed evidence records are admitted.

Regulatory And Risk

Standard	Title	Jurisdiction	Evidence Required
CE-UKCA	Electrical safety and UKCA/CE file	UK/EU	Technical file, electrical risk assessment and declarations.
WRAS	Water fittings compliance	UK	Approved fittings or evidence for water-contact components.

Hazard	S	O	D	RPN	Mitigation
Crop loss from pump failure	7	4	4	112	Dual pumps, flow monitoring and low-level alarms.
Electrical fault in humid grow area	8	3	4	96	Ingress-rated drivers, RCD protection and cable separation.

Section Issues And Score

bom

- blocker / critical_part_unpriced: Growing rack structure is critical but has no credible unit cost.
- blocker / critical_part_unpriced: LED grow lights is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Air circulation fans is critical but has no credible unit cost.
- blocker / critical_part_unpriced: CO2 sensor or dosing is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Reservoir and plumbing set is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Nutrient pump is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Environmental controller is critical but has no credible unit cost.
- blocker / critical_part_unpriced: RCD isolator panel is critical but has no credible unit cost.
- major / low_priced_line_ratio: Only 0% of BoM lines have prices.

Section Score Ledger

Rule: start at 9, subtract 3 per blocker, 1.5 per major, 0.5 per minor, floor at 1. Mean score 7.67. This is a deterministic gate score from current validators, not an external reviewer score or proof of real-world engineering quality. Reviewer acceptance is tracked separately in the verification ledger.

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
executive_summary	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
brief_requirements	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
design_modules	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
bom	9	8/1/0	-25.5	1	below target	blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); major/low_priced_line_ratio (-1.5)	Score is base minus deterministic deductions for attached section issues.
feasibility_notes	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
sources_references	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
Mean	7.67						